



KENYATTA UNIVERSITY
School of Agriculture and Environmental Sciences
Department of Agricultural Science and Technology

KRM 300/322: SOIL FERTILITY AND PLANT NUTRITION
PROF. JAYNE MUGWE

OBJECTIVES

At the end of course you are expected to be able to:

1. Describe basic concepts of soil fertility and plant nutrition
2. Explain concepts in soil fertility and how it influences plant growth and agricultural productivity
3. Discuss the various ways of managing soil fertility

COURSE OUTLINE

1. Introduction: Definitions, factors influencing soil fertility, status of African soils
2. Essential plant nutrients and their availability
3. Nitrogen nutrition
4. Phosphorus nutrition
5. Acid Soils
6. Soil Fertility Assessment
7. Nutrient uptake by plants – Soil plant relationships
8. Types and application methods of inorganic fertilizer
9. Organic fertilizers
10. Integrated soil fertility management

REFERENCE BOOKS Core reference materials

1. Amberger A (2006). Soil fertility and plant nutrition in the tropics and subtropics. IFA, Paris France
2. Brady NC (1990). The nature and properties of soils. Macmillan Publishing Company, New York. Tenth Edition

3. Eassh NS, Squer TJ, O'Dell D, Odoi E, Bratz MC (2015). Soil Science Simplified. 6th Edition. Willey Blackwell
4. Mugendi DN, Mucheru-Muna MW and Mugwe JN (Eds.) (2006). Soil Fertility: Enhancing Community Extension. Manilla Publishers, Nairobi, Kenya
5. Muñoz MA and Zornoza R (Eds) (2018). Soil Management and Climate Change: Effects on Organic Carbon, Nitrogen Dynamics, and Greenhouse Gas Emissions.
<https://www.sciencedirect.com/science/book/9780128121283>
6. Swift, M. J, Shepherd, K. D (Eds.) (2007). Saving Africa's soils: science and technology for improved soil management in Africa. [World Agroforestry Centre](#), Nairobi

Recommended reference material

1. Squer TJ, Norman J, Sivakumar MVK (Eds) (2011). Sustaining soil productivity in response to global climate change; Science, Policy and Ethics. Wiley Blackwell
2. Troeh TR and Thomson LM (2005). Soils and soil fertility. 6th Edition. Wiley Blackwell
3. Vanlauwe B, Diels J, Sanginga N and Merckx R (Eds.) (2002). Integrated Plant nutrient Management in Sub-Saharan Africa: From Concept to Practice. CAB International Wallingford, Oxon, UK.
4. Van Reuler, H., and Prins, W. (Eds.) (1993). The role of plant nutrients for sustainable food crop production in sub- Saharan Africa. VKP, Leidschendam, The Netherlands
5. Woomer PL, Swift MJ (eds) (1994). The biological management of tropical soil fertility. John Wiley and Sons, Newyork, USA.

Course Instructor

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Term paper assignment

- b) Write a term paper on the topic “Ways and Practices of managing soil fertility on smallholder farms in Sub-Saharan Africa”

Guidelines for term paper

For a good term paper, you have to read widely. Resources that can be used include books, journals and internet resources

The term paper should contain the following sections;

- Introduction (problem of soil fertility is discussed here)
- Body of the paper (major content is included here)
- Conclusions and recommendations:
- References

THE SUBMISSION SHOULD BE HANDWRITTEN

1: INTRODUCTION: BASICS, PRINCIPLES AND CONCEPTS

Learning Objectives

By the end of this topic, students will be able to:

1. Define key soil science and plant nutrition terms (e.g., soil, soil fertility, productivity, plant nutrition).
2. Explain the cause of loss in soil fertility
3. Discuss how various soil properties affect soil fertility
4. Explain soil fertility status in Africa
5. Discuss the causes of soil fertility decline in Africa

1.1 Soil and soil fertility

Introduction

Soil fertility and plant nutrition are vital for healthy crop growth. Fertile soil supplies plants with physical support, water, air, and essential nutrients. Understanding how soils and plants interact with nutrients helps improve crop productivity.

Key definitions

- **Soil:** The unconsolidated mineral material on the surface of the earth that serves as a natural medium for land plants (Soil Science Society of America, 1973)
- **Soil fertility:** refers to the inherent capacity of a soil to supply nutrients to plants in available forms, adequate amounts, and balanced proportions necessary for plant growth
- Two broad types of soils are Organic (> 20% organic matter) and Mineral soil (< 20% organic matter)
- **Plant nutrition:** refers to the interrelated steps by which a plant assimilates food and uses it for growth and replacement of tissue
- **Fertile soils:** Soil with the capacity to support a healthy crop. They have adequate soil moisture, nutrients, and capacity, and good soil workability. Fertile soils are formed from mineral-rich parent material under favorable climatic conditions.
- **Infertile soils:** Soils that lack the capacity to support a healthy crop. Caused by continuous cropping that depletes soil fertility
- **Soil productivity:** describes the ability of the soil to support crops when cultivated correctly, including supplements of nutrients from external sources to compensate for losses.
- **Available nutrient:** That portion of any element or compound in the soil that can readily be absorbed and assimilated by growing plants.
- **Soil health:** continued capacity of soil to function as a vital living ecosystem that sustains plants, animals, and humans
- **Broadcast:** Scatter seed or fertilizer on the surface of the soil
- **Carbon/nitrogen ratio:** The ratio of the weight of organic carbon (C) to the weight of total nitrogen (N) in a soil or in organic material.
- **Fertilizer:** Any organic and inorganic material of natural or synthetic origin added to a soil to supply certain elements essential to the growth of plants

- **Cation:** A positively charged ion.
- **Top-dressing:** An application of fertilizer to a soil after the crop stand has been established.
- **Cation exchange:** The interchange between a cation in a solution and another cation on the surface of any active material such as clay or organic.
- **Cation exchange capacity:** The sum of total of exchangeable cations that a soil can adsorb.

A **fertile soil** has the following properties:-

- Offer plants a sustainable and balanced supply of nutrients
- Transform nutrients into available form in soil solution
- Have a good water storage capacity and supply
- Contains soil organic matter that improves soil structure and soil moisture retention.
- Good soil structure, creating well-drained soil.
- Contains adequate amount of microorganisms that support plant growth.
- Often contains large amounts of topsoil.

Soil Fertility Losses

A loss of plant nutrients directly leads to soil infertility. Soil loses plant nutrients through: -

1. Crop removal in harvested farm produce and crop residues
2. Soil erosion- Soil containing plant nutrients is transported away by water or wind
3. Leaching- Plant nutrients are moved in solution form to deeper soil horizons by percolation
4. Volatilisation- Gaseous escape of nitrogen from the soil. This is very common when farmers leave heaps of exposed manure before application to the farms
5. Denitrification: Biochemical reduction of nitrates to gaseous forms of nitrogen
6. Fixation -Temporary conversion of plant nutrients into less available forms by chemical processes in the soil
7. Immobilization - Temporary conversion of plant nutrient from inorganic to organic forms making them not readily available for plant uptake

1.2 Soil characteristics influencing soil fertility

The most important soil characteristics that influence soil fertility include **soil texture, soil structure, soil moisture, soil aeration, plant nutrients, soil fauna and flora (micro-/macro-organisms), soil pH, soil depth, water holding capacity and humus content or soil organic matter.**

Soil Texture: Soil texture is arguably the single most important physical property of the soil in terms of soil fertility. It affects the soil's ability to hold water and nutrients. Sandy soils, with large particles, drain quickly and lose nutrients easily, making them less fertile. In contrast, clayey soils have fine particles that retain more water and nutrients but may hinder aeration and root growth if too compact. **Loamy soils**, which balance sand, silt, and clay, are generally the most fertile, as they combine good nutrient-holding capacity with proper drainage and aeration.

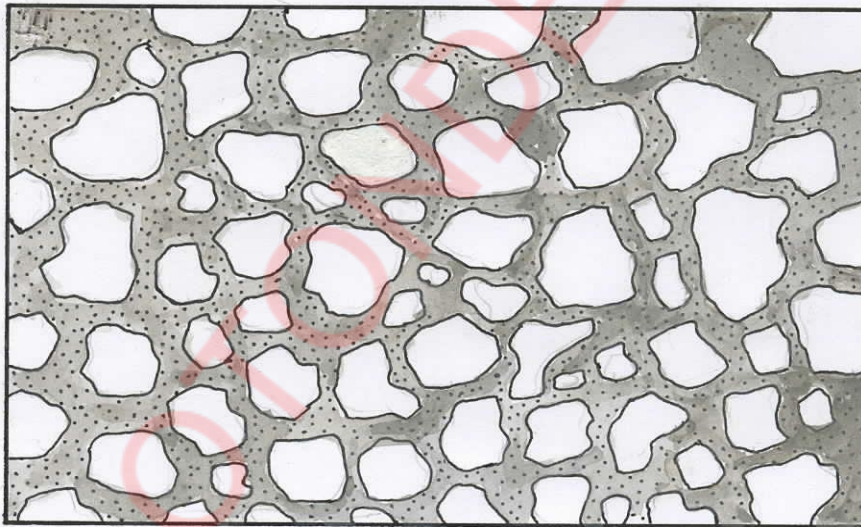
Soil structure: Soil structure refers to the arrangement of particles in a soil and how the individual soil particles clump or bind together. Structure is important because the arrangement of soil particles plays the biggest role in determining the size and shape of the pores that conduct air and water. It

also affects the plant's ability to send its roots through the soil. A poorly structured soil limits plant nutrients absorption and root penetration. A soil with good structure is allows root penetration and is important in utilization of nutrients by the crops.

Soil Nutrient Content: Consists of essential nutrient elements required for plant nutrition and metabolism. The behavior of plant nutrients in soil is very much connected with the clay fraction. The clay minerals in the soil are the ones mainly responsible for holding available soil nutrients in solution before they are absorbed by the plants.

Soil Moisture: Soil moisture greatly influences soil fertility because it controls the availability of water and nutrients to plants. Adequate moisture dissolves nutrients, making them easier for roots to absorb, while too little water limits nutrient uptake and plant growth. Excess moisture, however, can lead to waterlogging, poor aeration, and nutrient loss through leaching. Thus, balanced soil moisture is essential for maintaining soil fertility and healthy crop production.

Soil Aeration: A network of large pores is vital in allowing excess (gravitational) water to drain away and allow air circulation in the soil. The ease with which water flows through the soil is called **permeability** of the soil. If soil is impermeable, **water logging** occurs and air is excluded a situation very harmful to plant roots. A high water table can also cause water logging even when the soils are permeable.

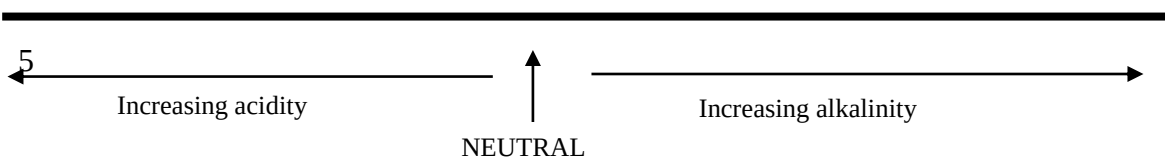


Soil constituents

Soil pH

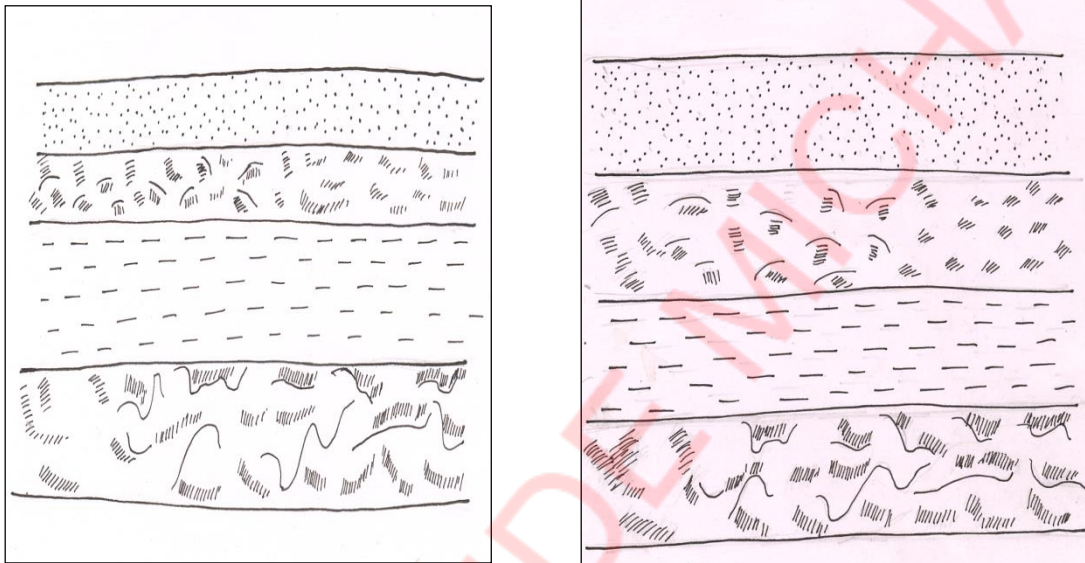
pH is defined as the negative logarithm of the hydrogen ions (H^+) concentration.

Soil PH: It is the indicator and regulator of soil chemical processes. Its measured following a scale of 1 to 14, acidity to alkalinity.



Soil pH because it plays an important role in plant growth. It affects crop production, soil chemistry, and microbial activity. It controls nutrient solubility and availability, with many essential nutrients **becoming unavailable if pH is too low or too high**. Acidic soils may reduce phosphorus, calcium, and magnesium uptake, limiting plant growth. Soil pH also influences microorganisms, strong acidity hinders decomposer bacteria, slowing organic matter breakdown and tying up nutrients like nitrogen.

Soil Depth: Determines the volume of soil accessible to the plant root system.



Soil profile of a shallow and a deep soil

Soil Organic Matter: **Soil organic matter (SOM)** greatly enhances soil fertility by improving both the physical and chemical properties of soil. It increases nutrient availability through mineralization, contributes to higher cation exchange capacity (CEC), and supplies essential nutrients like nitrogen, phosphorus, and sulfur. SOM also improves soil structure, which enhances water-holding capacity, aeration, and root penetration. In addition, it supports beneficial soil organisms that aid in nutrient cycling. Overall, soils with higher organic matter are more fertile, productive, and sustainable.

Cation Exchange Capacity (CEC): Cation exchange capacity (CEC) is the soil's ability to retain and supply nutrients to plants. High CEC soils, rich in clay and organic matter, hold nutrients better and are more fertile. In contrast, sandy soils have low CEC, so nutrients leach easily and fertility is reduced..

1.3 Soil Fertility Status in Africa

- Soils in Africa are inherently poor; they are developed from poor parent material, are old, and lack volcanic rejuvenation.
- About 65 % of the arable land in Africa is degraded due to water and wind erosion, loss of nutrients, physical deterioration, and salinization.
- The African continent loses a staggering US\$4 billion worth of soil nutrients annually due to erosion.
- Affects over 485 million people and threatens food security.

Soil fertility is variable among the countries

1. East Africa: widespread nutrient depletion on smallholder farms. Soil acidity is also a common problem in the highlands of Ethiopia, Rwanda, and Burundi.
2. Southern Africa: faces both nutrient-poor soils and soil erosion challenges. Salinization, particularly in semi-arid areas where irrigation is used.
3. North Africa: soil fertility is affected by the arid and semi-arid climate, leading to low organic matter content and high salinity levels. Intensive agriculture and over-irrigation have contributed to soil degradation, particularly in Egypt, Tunisia, and Algeria, where salinization is a significant problem
4. West and Central Africa: Soils in the rainforest regions of West and Central Africa, such as in the Democratic Republic of Congo and parts of Nigeria, are prone to **nutrient leaching due to high rainfall and poor management practices**

The **key causes of soil fertility decline in Africa** include the following;

1. **Population pressure:** As population density increases, the demand for agricultural land to produce food also rises. This often leads to converting marginal or previously uncultivated areas, such as sloping lands **into farmland**. Clearing forests for agriculture or fuelwood reduces organic matter inputs and disrupts natural nutrient cycling
2. **Nutrient depletion:** Nutrient mining is due to continuous cropping without adequate replenishment of nutrients. Soil erosion rates further accelerate nutrient depletion
3. **Inadequate nutrient management:** Africa has the lowest fertilizer use globally 10-15 kg of fertilizer applied per ha. Compared to over 200 kg in developed nations; Why?
 - a) High prices
 - b) Limited access
 - c) Inadequate distribution systems
 - d) Organics are not enough
4. **Increasing acidity:** Al toxicity; reduces the availability of nutrients (P, N). Received little attention; farmers use tolerant crops and animal manure (few use lime). What is the extent of acidic soils? Is the use of lime by farmers cost-effective? And on what crop?
5. **Poor land management practices:** Traditional systems, e.g., fallows and shifting cultivation, are no longer viable due to population pressure. Therefore there is extensive over-cultivation,, **intensive cultivation, Poor tillage practice, Burning agricultural residues**
6. **Climate change:** Extreme weather events; Frequent dry spells; Low adaptive capacity in SSA, Great impact on the organic matter and processes, Affect biomass production, decomposition rates, RISKY, WASTING RESOURCES

7. **Soil Erosion**

Water and wind erosion remove the fertile topsoil layer that contains the highest concentration of organic matter and nutrients. Poor land management, deforestation, and cultivation on steep slopes without conservation measures intensify erosion and degrade soil fertility.

1.4 Key initiative to address soil fertility in Africa

Key initiatives to address soil fertility in Africa

1. **National Programs:** Many African countries (e.g., Malawi, Ethiopia, Nigeria, Kenya) run fertilizer subsidy/voucher schemes, with Ethiopia also promoting soil testing and mapping.
2. **Africa Fertilizer & Soil Health Summit (2023, Nairobi):** Adopted the Nairobi Declaration; emphasized integrated soil fertility management (ISFM), climate-smart agriculture, and soil health monitoring.
3. **Abuja Declaration (2006):** AU Heads of State targeted raising fertilizer use from 8 kg/ha to 50 kg/ha by 2015; promoted subsidies, credit, and regional procurement.
4. **Alliance for A Green Revolution in Africa (AGRA):** Works with governments, the private sector, and farmers to improve seeds, soil fertility, and markets; promotes ISFM combining organic inputs, mineral fertilizers, and better agronomic practices.
5. **AU Agenda 2063:** Long-term framework to transform Africa; prioritizes investment in agriculture and soil fertility to achieve food security and economic growth.

CHAPTER 2: ESSENTIAL PLANT NUTRIENTS

Learning Objectives

By the end of this chapter, students should be able to:

1. Define what makes a nutrient “essential” for plant growth.
2. List all the essential plant nutrients and classify them as macronutrients and micronutrients.
3. Describe the roles/functions of each essential nutrient in plant physiology and metabolism.
4. Identify the deficiency and toxicity symptoms of major nutrients in crops.
5. Explain yield response to nutrient application.
6. Discuss the concept of nutrient use efficiency

2.1 Essential plant nutrients and their roles

Plants obtain three essential elements: carbon, hydrogen, and oxygen from air and water. These elements comprise about 94% or more of a plant’s dry matter and are vital for growth. The remaining essential nutrients, though accounting for less than 6% of plant dry matter, are critical for crop production. Deficiencies in one or more of these elements often limit plant growth and reduce yields.

These essential elements are generally divided into two groups (Table 1).

- i) **The macronutrients** are nitrogen (N), phosphorus (P), potassium (K), sulfur (S), calcium (Ca), and magnesium (Mg).
- ii) **The micronutrients** (or sometimes ‘minor’ or ‘trace’ elements) because those elements are required in small (micro) amounts by plants. They include manganese (Mn), iron (Fe), boron (B), zinc (Zn), copper (Cu), molybdenum (Mo), chlorine (Cl), cobalt (Co), nickel (Ni), sodium (Na) and silicon (Si).

Why essential?

- The plant cannot complete its life cycle in absence of any of them
- Deficiency symptoms of any of them can only be corrected by its supply. They cannot be substituted
- Deficiency of any one of them can cause abnormal or restricted plant growth
- Each of them is directly involved in plant metabolism besides any other functions

Macro nutrients: Carbon (C), hydrogen (H), oxygen (O), nitrogen (N), phosphorus (P), potassium (K), calcium (Ca), magnesium (Mg), and sulphur (S). They are;

- Constituents of many plant components such as nucleic acids, protein and chlorophyll
- Essential processes such as energy transfer, maintenance of internal pressure and enzyme functions require them
- Required by plants in large amounts
- Added to the soil mainly through soil-applied fertilizers (N, P, K, Ca and S)

Micro-nutrients: Boron (B), chlorine (Cl), iron (Fe), zinc (Zn), copper (Cu), molybdenum (Mo), and manganese (Mn). They;

- Play essential functions in plant metabolism
- Required by plants in small amounts
- Play critical roles as **enzyme activators**, in **hormone regulation**, and in processes like **photosynthesis, nitrogen fixation, and reproductive development**. Deficiencies, even in tiny amounts, can cause **serious physiological disorders** in plants, leading to stunted growth, poor yield, or low quality produce.
- Unlike macronutrients, which are often supplied through bulk fertilizers, micronutrients are usually applied in **small doses as foliar sprays, soil amendments, or seed treatments**.

Table 1: Essential elements for plants

	From air and water	
Element	Major or macro	Symbol
Carbon		C
Hydrogen		H
Oxygen		O
Nitrogen		N
	From soil and fertilizers	
Nitrogen	Major or macro	N
Phosphorus	“	P
Potassium	“	K
Calcium	“	Ca
Magnesium	“	Mg
Sulfur	“	S
Iron	Micronutrient	Fe
Boron	“	B
Manganese	“	Mn
Copper	“	Cu
Zinc	“	Zn
Molybdenum	“	Mo
Copper	“	Cl
Cobalt	“	Co

Table 1: Roles of essential plant Nutrients

Nutrient	Category	Key Roles in Plants
Carbon (C)	Macronutrient (Non-mineral)	Main component of carbohydrates, proteins, fats, cell walls (from CO ₂ via photosynthesis).
Hydrogen (H)	Macronutrient (Non-mineral)	Part of water and organic compounds; involved in photosynthesis and energy transfer.
Oxygen (O)	Macronutrient (Non-mineral)	Essential for respiration and structure of organic molecules.
Nitrogen (N)	Macronutrient	Component of amino acids, proteins, nucleic acids, chlorophyll; promotes vegetative growth.
Phosphorus (P)	Macronutrient	Part of ATP, nucleic acids, phospholipids; vital for energy transfer, root development, flowering.
Potassium (K)	Macronutrient	Regulates enzyme activation, osmoregulation, stomatal opening/closing; improves drought tolerance and quality of fruits/grains.
Calcium (Ca)	Secondary Macronutrient	Builds cell walls (calcium pectate), strengthens stems, aids in root and meristem development.
Magnesium (Mg)	Secondary Macronutrient	Central atom in chlorophyll; activates enzymes; important in photosynthesis and energy metabolism.
Sulfur (S)	Secondary Macronutrient	Component of some amino acids (cystine, methionine), vitamins, and enzymes; promotes seed and oil formation.
Iron (Fe)	Micronutrient	Involved in chlorophyll synthesis, electron transport in photosynthesis and respiration.
Manganese (Mn)	Micronutrient	Activates enzymes for photosynthesis, respiration, and nitrogen metabolism.
Zinc (Zn)	Micronutrient	Needed for auxin (growth hormone) synthesis, enzyme activation, and protein formation.
Copper (Cu)	Micronutrient	Part of enzymes for photosynthesis, lignin formation, and reproductive growth.
Boron (B)	Micronutrient	Important for cell wall formation, pollen tube growth, flowering, and fruit/seed development.
Molybdenum (Mo)	Micronutrient	Required for enzymes in nitrogen fixation and nitrate reduction.
Chlorine (Cl)	Micronutrient	Involved in photosynthesis (oxygen-evolving complex), osmotic and ionic balance.
Nickel (Ni)	Micronutrient	Required for enzyme urease activity, enabling plants to use urea nitrogen.

MACRONUTRIENTS			MICRONUTRIENTS
<i>Non-Mineral</i>	<i>Primary</i>	<i>Secondary</i>	
Mostly from air and water	Mostly from soils	Mostly from soils	Mostly from soils
Carbon	Nitrogen	Calcium	Iron
Hydrogen	Phosphorus	Magnesium	Manganese
Oxygen	Potassium	Sulfur	Boron
			Zinc
			Copper
			Molybdenum
			Chlorine
			Cobalt
			Nickel

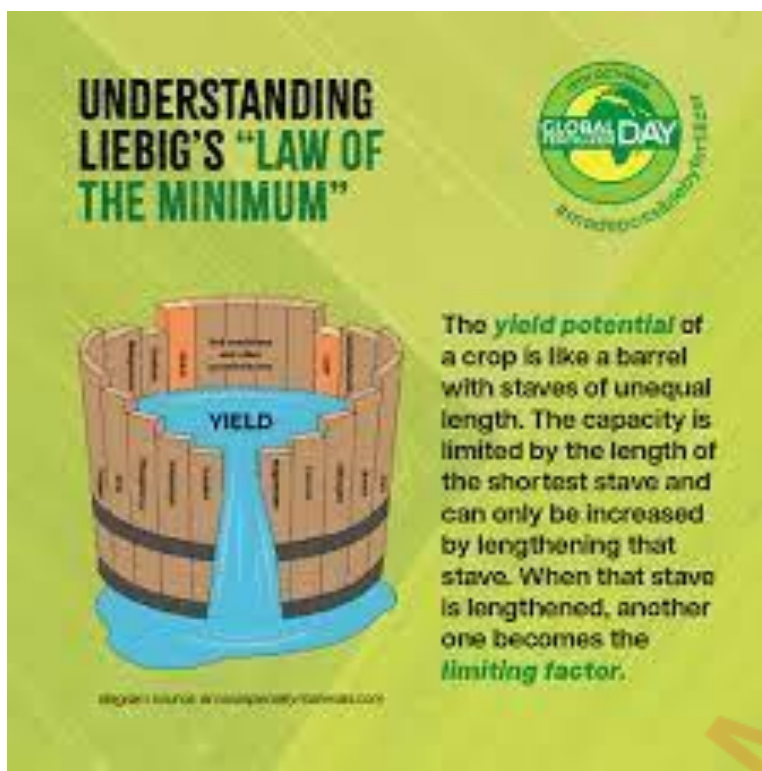
2.2 Yield response to nutrient supply

Definition: Yield response to nutrient supply refers to the **relationship between the amount of nutrients available in the soil and the crop yield produced**. It shows how crop growth and productivity change as nutrient levels increase

Two key principles guide the yield response concept

1. The Concept of Liebig's Law of the Minimum):

This concept states that crop growth is limited by the nutrient in shortest supply, even if all other nutrients are adequate. Adding nutrients that are not limiting will not significantly increase yield. Example: If nitrogen is deficient, adding phosphorus or potassium will not improve growth until nitrogen is supplied.

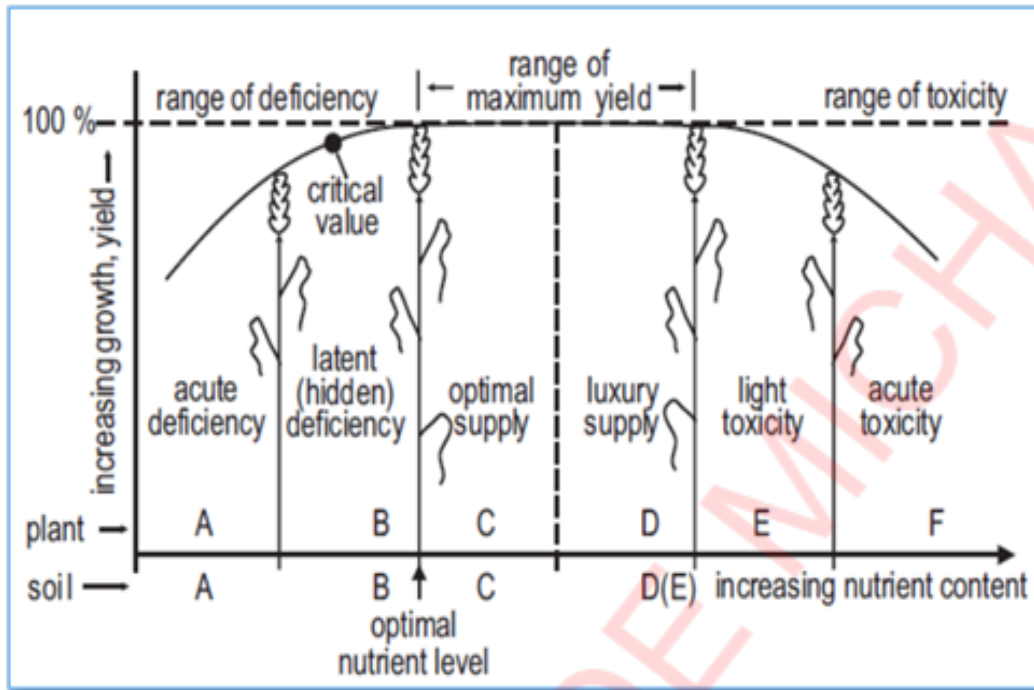


2. Yield response curve

Nutrients are not only limited and costly to replenish, but improper application can have serious consequences for plant health and yield. Plant response to nutrient levels spans a continuum, from **acute deficiency** to **acute toxicity**. This range can be broadly categorized into three main groups: deficient, optimal, and excess. For more precise evaluation, these groups can be further subdivided into six distinct ranges. This include:

- i. **Acute deficiency:** the plant **lacks** essential nutrients for basic growth functions. Symptoms manifest quickly and yield is severely suppressed.
- ii. **Marginal or latent deficiency (hidden hunger):** Nutrient levels are still insufficient, but less extreme than acute deficiency. Plants may survive but exhibit poor growth, lower vigor, and visible symptoms of stress. Yields will be suboptimal, and the plant's ability to resist pests or environmental stress is reduced
- iii. **Optimal supply:** target range where nutrient levels are sufficient to support healthy plant growth. The plant can express its full genetic potential in terms of yield and quality. Symptoms of stress are absent, and nutrient use efficiency is maximized. This range ensures the best possible yield for the inputs provided.
- iv. **Luxury supply:** the plant absorbs more nutrients than needed, but without immediate harm. Yield does not increase beyond what is achieved in the sufficiency range. However, the excess nutrients could be stored in plant tissues, potentially leading to inefficiencies and wastage.
- v. **Marginal or light (hidden) toxicity:** Nutrient levels become excessive, beginning to inhibit plant growth. Symptoms such as leaf burn, wilting, and poor root development may appear. Yields start to decline, and plant health deteriorates as the excess nutrients disrupt physiological processes.
- vi. **Acute toxicity:** excess nutrient levels become highly damaging to the plant. Severe toxicity can cause plant death or significant yield losses due to cellular damage, nutrient imbalances, and physiological stress. Symptoms are pronounced and irreversible.

By understanding these six ranges, we can better manage nutrient applications to ensure plants remain within the optimal range, thereby maximizing yield and preventing the negative effects of deficiency or toxicity.



Yield response curve showing yield response to nutrient supply Finck (1992)

Critical Range concept

Critical range refers to the nutrient supply range where growth and yield transition from being limited by deficiency to being adequate for optimal plant performance. The critical range serves as a target for nutrient application.

2.3 Nutrient Use Efficiency

Nutrient use efficiency (NUE) refers to the **effectiveness with which plants utilize available nutrients for growth, development, and yield production**. Improving NUE is essential for sustainable agriculture, as it can lead to higher crop yields with lower inputs, reducing the environmental impact of fertilization.

One of the most important terms used to express nutrient use efficiency is Agronomic efficiency. **Agronomic Use Efficiency (AUE)** refers to the increase in crop yield obtained per unit of nutrient applied, typically measured for fertilizers like nitrogen, phosphorus, or potassium. It evaluates how efficiently plants convert applied nutrients into harvestable products, such as grain or biomass.

Calculation

: AUE is the ratio of the yield increase (due to fertilizer application) to the amount of nutrient applied. It is expressed as:

$$(AUE) = \frac{\text{Crop Yield with Fertilizer (kg/ha)} - \text{Crop Yield without Fertilizer (kg/ha)}}{\text{Amount of Fertilizer Applied (kg/ha)}}$$

- High NUE means a larger proportion of applied nutrients is taken up and utilized by the crop; low NUE indicates nutrient losses to the environment.

Importance of optimizing NUE:

1. Optimizing NUE is critical for **maximizing crop productivity** while minimizing **fertilizer costs**.
2. Improves **profitability** for farmers and reduces **negative environmental impacts** such as nutrient leaching, runoff, and greenhouse gas emissions.

How to improve nutrient use efficiency

Use of 4 R principle

- Right Source: Use fertilizers with nutrients in forms readily available to crops (e.g., ammonium nitrate, urea).
- Right Rate: Apply nutrients according to soil tests and crop needs, avoiding over- or under-application.
- Right Time: Synchronize nutrient application with critical crop growth stages for maximum uptake.
- Right Place (Placement): Place nutrients where roots can access them efficiently, e.g., banding or localized application.
- Integrated Nutrient Management (INM): Combine organic amendments, mineral fertilizers, and biofertilizers to enhance nutrient availability.
- Soil and Water Management: Improve soil structure, moisture retention, and reduce erosion to minimize nutrient losses.

CHAPTER 3: NITROGEN NUTRITION

Learning objectives

By the end of this chapter, students should be able to:

1. Describe the role of nitrogen in plant growth, metabolism, and crop productivity.
2. Identify and explain the major forms of nitrogen present in soils and how plants absorb them.
3. Illustrate the nitrogen cycle, highlighting key processes
4. Explain the Nitrogen leaching problem
5. Differentiate between symbiotic and non-symbiotic biological nitrogen fixation (BNF)
6. Discuss factors that influence nitrogen availability and BNF efficiency, including soil pH, moisture, temperature, nutrient levels, and salinity

3.1 Importance and availability

Nitrogen (N) is one of the most important plant nutrient and is the most frequently deficient.. This is despite the fact that most of the air we breathe is N_2 . This is because most of atmospheric N is unavailable for use by organisms. It is for this reason that the productivity of most agricultural crops is mainly limited by the inadequate availability of N.

The content of surface soils ranges from 0.02 to 0.5%, the majority of which is from soil organic matter (SOM), **supplying about 95-99%**. The purpose of this lesson is therefore, to make you appreciate the role N plays in plant growth and yield.

Forms of N taken up by the plant

Plant roots take up nitrogen from the soil solution principally as **nitrate (NO_3^-)** and **ammonium ions (NH_4^+)**. The nitrite is also available for uptake, but once formed, it is short-lived.

Nitrates: Produced by **nitrification** of ammonium by soil bacteria (*Nitrosomonas* $\rightarrow NO_2^- \rightarrow$ *Nitrobacter* $\rightarrow NO_3^-$). Highly **soluble and mobile** in soil water; easily leached in sandy soils. Once inside the plant, nitrate is reduced to ammonium in the roots or leaves by enzymes

Influence of N on plant growth

Roles in the plant

Nitrogen is an integral component of many essential plant compounds and required by plants in the largest amounts. It is a major part of all amino acids, the building blocks of proteins including the enzymes. It is also present in the bases that make up nucleic acids such as DNA (deoxyribonucleic acid) and RNA (ribonucleic acid)

In plants, much of the nitrogen is used in chlorophyll molecules which are essential for photosynthesis and further growth. Nitrogen promotes rapid growth, increases leaf size and

quality, hastens crop maturity, and promotes fruit and seed development. N also encourages aboveground vegetative growth and gives leaves a deep green colour. A healthy plant foliage generally contains 2.5 to 4 % N

Oversupply of N

Excessive supply have the following harmful effects;

1. **Excessive vegetative growth** occurs, and plants become prone to falling over (lodging)
2. **Delayed maturity:** High N applications may delay maturity
3. **Susceptibility to pests and diseases:** In some cases excessive supply may make a plant more susceptible to disease and pest attack
4. **Adverse effects on fruits:** Colour and flavor of fruits may be poor
5. **Poor fruit production:** Flower production in ornamentals is reduced in favor of over abundant foliage

Deficiency

- **Stunted growth:** Nitrogen-deficient plants exhibit slow stunted growth
- **Chlorosis:** The leaves have a pale yellowish green color (chlorosis). This is because N is an integral part of chlorophyll, which is the primary absorber of light energy needed for photosynthesis, and adequate supply is associated with high photosynthetic activity. deficiency symptoms generally appear on the bottom leaves first. In severe cases, the lower leaves have a “fired” appearance on the tips, turn brown, usually disintegrate, and fall off.
- **Low yields and quality:** In leafy crops such as tobacco, vegetables, forage and pasture crops, low nitrogen results in low yield and quality. When grain crops, such as corn and small grains, are deficient, they generally exhibit yellow leaf tips, stunted growth and low yields of poor-quality grain
- **Low protein content:** In a nitrogen-deficient plant, the protein content is low. Eg forage growing in N-deficient soils will have low crude protein
- **Low shoot-to-root ratio:** due to inadequate growth of above-ground biomass

3.2 The Nitrogen Cycle

The movement of nitrogen between the atmosphere, biosphere, and geosphere in different forms is described as the nitrogen cycle. It is one of the major biogeochemical cycles. Earth's atmosphere is approximately 78-80% nitrogen making it the largest pool of nitrogen.

Nutrient cycling is the continuous movement of nutrients between soil and plants, mediated by soil organisms that decompose plant residues and release nutrients for plant uptake. Natural forest ecosystems have efficient, closed nutrient cycles, while agricultural systems often experience higher nutrient losses.

The nitrogen cycle is especially important, converting atmospheric nitrogen into plant-available forms through fixation, supporting plant growth, maintaining soil fertility, and regulating environmental nitrogen levels to prevent pollution and eutrophication.

The Processes of the nitrogen cycle

Five main processes cycle nitrogen through the biosphere, atmosphere, and geosphere: (Figure 2.1),

1. Nitrogen fixation
2. Assimilation: Nitrogen uptake (organismal growth),
3. Nitrogen mineralization (decay),
4. Nitrification
5. Dentrification

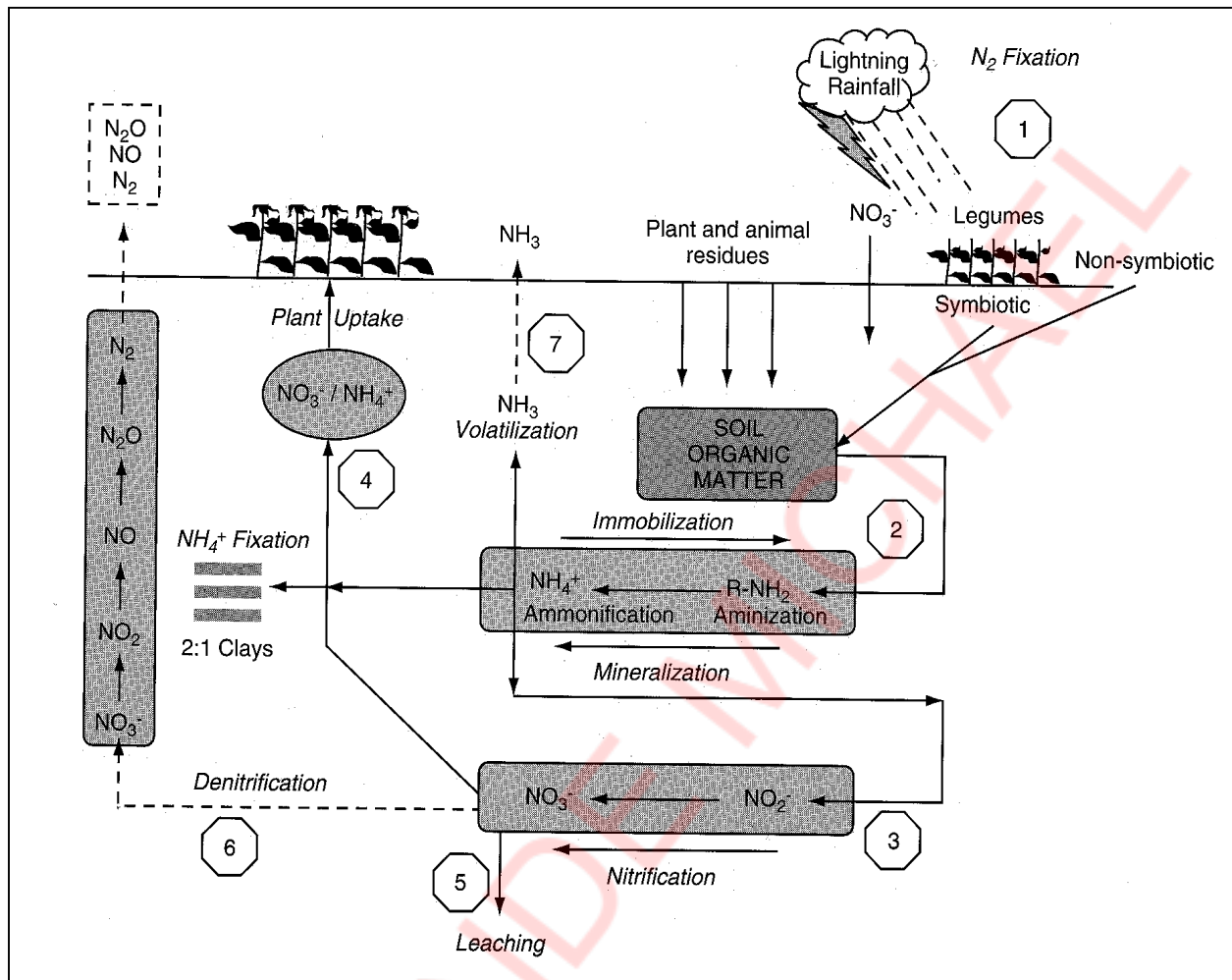


Figure 2.1: The Nitrogen Cycle

1. Nitrogen fixation

Three fixation mechanisms are important;

a) Biological nitrogen fixation (BNF):

BNF converts atmospheric N gas to nitrogen-containing compounds that become available to plants. Some **symbiotic** bacteria (most often associated with leguminous plants) and some free-living bacteria are able to fix nitrogen as organic nitrogen. There are also nitrogen-fixing bacteria that exist without plant hosts, known as free-living nitrogen fixers. In aquatic environments, blue-green algae (really a bacteria called cyanobacteria) is an important free-living nitrogen fixer.

b) Industrial N-fixation:

Under great pressure, at a temperature of 600°C , and with the use of a catalyst, atmospheric nitrogen and hydrogen (usually derived from natural gas or petroleum) can be combined to form ammonia (NH_3). In the Haber-Bosch process, N_2 is converted together with hydrogen gas (H_2) into ammonia (NH_3) which is used to make fertilizer and explosives.

Ammonia can be used directly as fertilizer, but most of it is further processed to urea and ammonium nitrate

c) Atmospheric fixation

The enormous energy of lightening breaks nitrogen molecules and enables their atoms to combine with oxygen atoms in the air forming nitrogen oxides. These dissolve in rain, forming nitrates that are carried to the earth. Atmospheric N fixation probably contributes 50% of the total N fixed

2. Assimilation

Nitrogen assimilation is the process in which inorganic nitrogen (NH_4^+ or NO_3^-) is converted into organic nitrogen forms. Crops are harvested, and nitrogen is removed from the field in the harvested plant material. This represents the direct transfer of soil nitrogen into plant biomass. And Ensures plant growth, higher yields, and contributes organic matter back to the soil after plant residues decompose.

3. N Mineralisation or Ammonification

When a plant dies, an animal dies, or an animal expels waste, the initial form of nitrogen is organic. The conversion of organically bound N into inorganic mineral forms (nitrates and ammonium (NH_4)) is termed as ammonification or mineralization. Process if mineralization is aided by a variety of organism. The rate depends largely on the temperature, moisture and aeration status of the soil

Maximum rates occur in hot climates and well-aerated soils. Immobilization is the opposite of mineralization. It is the conversion of inorganic NO_3^- and NH_4^+ into organic form. If decomposing OM contains low N relative to C, microorganism will immobilize NH_4^+ and NO_3^- in the soil. This is because when organic residues are being decomposed in the soil, the growth of microorganisms may require more N than that released. The organism then take up mineral N and therefore make it unavailable. An example is incorporating maize stover- a period of immobilization is experienced and the crop may turn yellow.

4. Nitrification

The conversion of NH_4^+ to nitrates is performed primarily by soil-living bacteria and other nitrifying bacteria. Nitrification is a two-stage process in which NH_4^+ is first converted to NO_2^- and then to NO_3^- . The first stage is accomplished by Nitrosomonas species, which converts ammonia to nitrites (NO_2^-) while the second stage is performed by nitrobacter. It is important for the nitrites to be converted to nitrates because accumulated nitrites are toxic to plant life.

5. Denitrification

- Denitrification is the reduction of nitrates back into the largely inert nitrogen gas (N₂), completing the nitrogen cycle.
- When soils become waterlogged oxygen is excluded and anaerobic decomposition takes place converting nitrates into nitrogen gases (NO, N₂O, N₂) and returned to the atmosphere.
- This is common in flooded soils such as those found in natural wetlands or rice paddies

3.3 The C:N ratio

The C:N ratio of organic matter means the amount of carbon relative to the amount of nitrogen present. The C:N ratio of organic matter is important because of what happens when organic matter is incorporated into soils. Generally there is always more carbon than nitrogen in organic matter. If decomposing OM contains low N relative to C, i.e. high C: N ratio microorganism will immobilize NH₄ and NO₃ in the soil. This is because when organic residues are being decomposed in the soil, the growth of microorganism may require more N than that released. The organism then takes up mineral N and therefore makes it unavailable known as immobilization. Example is incorporating maize stover or wheat- a period of immobilization is experienced and crop may turn yellow in the short term.

If materials added contain much N in proportion to the C present e.g. manure, tithonia, mucuna) there will be no decrease of mineral N but it will increase instead.

Quality of manures in smallholder farms of Africa are variable; majority are of low quality due to poor management (N range 0.5 to 2.0%). Poorly decomposed manure has high C:N ratio: recommendation is to use well decomposed manure with Appropriate C:N ratio 15:1 upto 35:1:

Materials with low C:N ratio eg tithonia tend to decompose very fast and vice versa

The N content of OM ranges from (1.5 to 5.5). while C ranges from 45 to 58%

Below are the average C:N ratios for some common organic materials . The materials containing high amounts of carbon are considered "browns," and materials containing high amounts of nitrogen are considered "greens."

Estimated Carbon-to-Nitrogen Ratios	
Browns = High Carbon	C:N
Ashes, wood	25:1
Cardboard, shredded	350:1
Corn stalks	75:1
Fruit waste	35:1
Leaves	60:1

Newspaper, shredded	175:1
Peanut shells	35:1
Pine needles	80:1
Sawdust	325:1
Straw	75:1
Wood chips	400:1
Greens = High Nitrogen	C:N
Alfalfa	12:1
Clover	23:1
Coffee grounds	20:1
Food waste	20:1
Garden waste	30:1
Grass clippings	20:1
Hay	25:1
Manures	15:1
Seaweed	19:1
Vegetable scraps	25:1
Weeds	30:1

3.4 The nitrogen leaching problem

Nitrogen leaching is a significant soil fertility and environmental issue caused mainly by the mobility of nitrate in soils

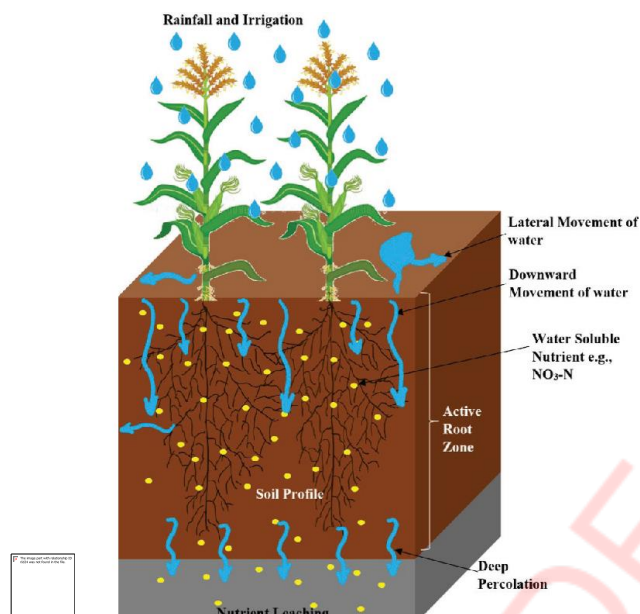
In contrast to ammonium ions, which carry a positive charge, the negatively charged nitrate ions are not adsorbed by the negatively charged colloids that dominate most soils. Therefore, nitrate ions moves downward freely with drainage water, and are thus easily leached from the soil. This loss is of concern for 2 reasons; **it depletes the ecosystem and also causes several serious environmental problems.**

Nitrate lost by leaching is generally carried by drainage waters to the ground water. It may reach domestic wells, rivers, lakes etc. Nitrates in drinking water is a health hazard. It can cause **eutrophication**, a process leading to high algal, especially blue-green algal populations and the death of aquatic life due to excessive demand for oxygen

Leaching is minimal in semi-arid areas and very sandy soils are particularly susceptible to leaching losses.

Avoid excess nitrogen fertilizer application: When fertilization is too high, exceeding plant requirements, excess nitrates is prone to leaching eg in developed nations

Timing is important. Potential for leaching is high where rainfall is high and plant use is low eg early in the season. Careful soil management should avoid applying excesses and applying modest rates when needed- split application



Why NO_3^- is highly leached

Soil particles like clay and organic matter carry a negative charge.

Positively charged nutrients (cations) such as ammonium (NH_4^+), calcium (Ca^{2+}), and potassium (K^+) are held on the soil's cation exchange sites.

Nitrate is negatively charged, so it is repelled by the soil particles and remains dissolved in the soil water. This means it is not retained by the soil matrix and moves freely with

3.5 Biological nitrogen fixation

BNF occurs through a number of microorganism systems with or without direct association with higher plants. The legume-bacteria systems have received most attention but other systems are also important

a) Symbiotic fixation with legumes

Legumes and bacteria' symbiosis (mutually beneficial relationship) provide the major biological source of fixed N in the soil. These are Rhizobium and Bradyrhizobium, and live in the root nodules of legumes (such as peas or beans). Rhizobium is fast growing, while Bradyrhizobium is slow growing. They infect the root hairs, forming root nodules that serve as sites for N fixation. In this relationship, the host plant supplies the bacteria with carbohydrates for energy and the bacteria supply the plant with nitrogen-fixing compounds. Nutrient-poor soils can be planted with legumes to enrich them with nitrogen.

The symbiotic N fixation is beneficial to both forestry and agriculture. Examples of crops/legumes with associations are cowpeas (*Vigna unguiculata*), Pea nut (*Arachis*), Soybean (*Glycine max*), pigeon peas (*Cajanus cajan*), Bean (*Phaseolus vulgaris*)

Legume trees examples *Leucaena trichandra*, *Gliricidia sepium*, *Callindra calothyrsus*, *Senna siamea* N fixation by perennial legumes is higher than that fixed by annuals

The rhizobium are specific- meaning bacterial that live symbiotically with soybean will not fix in peanuts. Inoculation of the legumes seed with the correct inoculum is recommended. Eg soybean

Nitrogen availability to non-legumes

Nitrogen fixed in legume root nodules benefits the host plant, resulting in protein-rich tissues and grains. Additionally, part of the fixed nitrogen becomes available to nearby or subsequent non-legume crops, often improving their yields due to enhanced soil nitrogen availability.

Research has identified herbaceous legumes such as *Mucuna pruriens* and *Crotalaria ochroleuca* as effective in improving soil fertility through intercropping or crop rotation. In agroforestry, trees like *Sesbania sesban*, *Tephrosia vogelii*, and *Crotalaria grahamiana* are commonly used in improved fallows to enrich soil nitrogen. However, intercropping with common beans often contributes little nitrogen because beans are relatively weak nitrogen fixers.

By incorporating these legumes into cropping systems, farmers can improve nitrogen availability for non-legume crops, thus enhancing soil fertility and crop productivity.

Non-Symbiotic Nitrogen Fixation:

Non-symbiotic nitrogen fixation is carried out by **free-living microorganisms** (e.g., *Azotobacter*, *Beijerinckia*, *Clostridium*, cyanobacteria like *Anabaena* and *Nostoc*) that convert atmospheric nitrogen (N_2) into ammonium **without forming nodules on plants**. They use **nitrogenase enzyme** and require energy from soil organic matter.

Although they fix **less nitrogen (5–20 kg N/ha/year)** than legumes, they still improve soil fertility—especially in **rice paddies** and organic-rich soils.

Their activity depends on soil **moisture, pH, temperature, aeration, and organic matter content**.

3.6 Factors limiting biological nitrogen fixation

Three main factors limit BNF: 1. Edaphic, 2. Climatic and 3. Biotic

Edaphic Factors: Edaphic factors relate to the soil. The six main edaphic factors limiting biological nitrogen fixation are:

- i. Excessive soil moisture
- ii. Drought

- iii. Soil acidity
- iv. P deficiency
- v. Excess mineral N
- vi. Deficiency of Ca, Mo and B.
- vii. Soil salinity

Excessive moisture and waterlogging prevent the development of root hair and sites of nodulation, and interfere with a normal diffusion of O₂ in the root system of plants.

Drought reduces the number of rhizobia in soils, and inhibits nodulation and N₂ fixation. Prolonged drought will promote nodule decay.

Soil acidity and related problems of Ca deficiency and aluminum and manganese toxicity adversely affect nodulation, N₂ fixation and plant growth. Research work on the identification of symbioses adapted to acid soil should focus on the host plant, because effective rhizobia adapted to soil acidity can be found naturally and can be produced through genetic manipulations.

Phosphorus deficiency is common in tropical Africa and reduces nodulation, N₂ fixation and plant growth. Identification of plant species adapted to low-P soils is a good strategy to overcome this soil constraint. Dual inoculation with effective rhizobia and mycorrhizal fungi shows synergistic effects on nodulation and N₂ fixation in low P soils. The use of local rock phosphate has been recommended, particularly in acid soils, as an inexpensive source of P.

Mineral N inhibits the *Rhizobium* infection process and also inhibits N₂ fixation. The former problem probably results from impairment of the recognition mechanisms by nitrates, while the latter is probably due to diversion of photosynthates toward assimilation of nitrates. Some strains of *Rhizobium*, and particularly stem-nodulating *Azorhizobium caulinodans*, fix N₂ actively even when plants are growing in high-N soils (e.g., in the presence of 200 kg fertilizer N ha⁻¹). Application of large quantities of fertilizer N inhibits N₂ fixation, but low doses (<30 kg N ha⁻¹) of fertilizer N can stimulate early growth of legumes and increase their overall N₂ fixation. The amount of this starter N must be defined in relation to available soil N.

Deficiency of various microelements (Cu, Mo, Co, B) are necessary for N₂ fixation. Some of these are components of nitrogenase for example Mo.

Soil Salinity

- Soil salinity; High salt levels reduce plant growth, disturb water uptake, and inhibit rhizobia survival and nitrogenase enzyme activity.

Climatic Factors: The two important climatic determinants affecting BNF are temperature and light.

Extreme temperatures affect N₂ fixation adversely. This is easy to understand because N₂ fixation is an enzymatic process. However, there are differences between symbiotic systems in their ability to tolerate high (>35°C) and low (<25°C) temperatures.

Biotic Factors: The main biotic factor is the presence or absence of the required rhizobia species, which constitutes the major constraint in the nitrogen fixation process. The other limiting biotic factors could be:

- ☐ excessive defoliation of host plant,
- ☐ crop competition, and
- ☐ insects and nematodes

Inoculation of legumes

In soils where specific and effective rhizobia are absent or present in low numbers, inoculating the soil with rhizobia is necessary to ensure proper nodulation and nitrogen fixation. This process, known as inoculation, introduces the required rhizobia to promote efficient N-fixing activity in legume roots. However, if sufficient numbers of effective rhizobia are already present in the soil, inoculation may not be required.

In Kenya use of Biofix is one of the ways of increasing rhizobium in the soil. We call it inoculate.



Biofix is designed for leguminous crops, including:

Field Crops: Beans, cowpeas, groundnuts, soybeans.

Pasture Crops: Lucerne, desmodium.

Trees: Sesbania, calliandra. It promotes nodulation, enabling plants to convert atmospheric nitrogen into usable nutrients, improving growth and yields.

- To check if inoculation is needed, observe the color of root nodules on legumes during mid-season. Healthy, nitrogen-fixing nodules are **pink or red** inside due to leghemoglobin.
- If most nodules are **white, grey, or green**, nitrogen fixation is poor—often caused by ineffective rhizobia strains, nutrient deficiencies, pod-filling, or plant stress—indicating that inoculation with an effective strain may be necessary. Are there few or no nodules on a young, uninoculated legume growing in low-N soil? This usually coincides with nitrogen deficiency symptoms in the plant.

Once introduced, inoculant rhizobia can persist in the soil for long periods, especially when the host plant is grown regularly or permanently. This persistence can reduce the need for re-inoculation in subsequent seasons, provided the inoculant strains maintain their effectiveness over time. Inoculation is typically recommended for newly introduced legumes.



SIR OTONDE MICHAEL

CHAPTER 4: PHOSPHORUS NUTRITION

Learning Objectives

By the end of this chapter, students should be able to:

1. Explain the importance of phosphorus in plant growth and development.
2. Describe the forms of phosphorus present in soils (organic and inorganic).
3. Understand the phosphorus cycle in soil–plant systems.
4. Identify the factors affecting phosphorus availability in soils (e.g., pH, fixation, organic matter).
5. Discuss ways of enhancing phosphorus availability on smallholder farms
6. List common phosphorus fertilizers and their properties

4.1 Availability of phosphorus in soil

This lesson discusses the importance of phosphorus, another important plant nutrient, in plant nutrition. Phosphorus is mainly limiting in soil due to three reasons: 1) **Low total P amount**, 2) **unavailable forms** and 3) **P fixation**.

P is also not supplied through fixation processes and hence it has to be supplied to the soil via inorganic forms or fertilizers. The purpose of this lesson is to make you understand P nutrition in plants and how to manage shortage of P in the soil. It is important you know that; many phosphate compounds are not very soluble in water; therefore, most of the phosphate in natural systems exists in solid form which may explain the general low P in soil solution.

Uptake of Phosphorus by plants

Phosphorus in the soil occurs in two main forms: organic and inorganic. **Organic phosphorus**, which makes up about 30–65% of total soil phosphorus, is found in organic matter such as plant residues and microbial biomass. It becomes available to plants through mineralization by soil microbes. The remaining 35–70% exists in **inorganic forms**, largely bound to calcium, iron, or aluminum compounds. These inorganic phosphates dissolve in soil water to release phosphate ions that plants can absorb.

Plant roots take up phosphorus only in the dissolved form from the soil solution, mainly as the phosphate ions H_2PO_4^- and HPO_4^{2-} . The dominant ion depends on soil pH: in strongly acidic soils (pH 4–5.5), the monovalent H_2PO_4^- form dominates, whereas in neutral to alkaline soils (pH above 7), the divalent HPO_4^{2-} form prevails. This relationship means that soil pH plays a critical role in determining phosphorus availability to crops.

The role of P in plant nutrition

As we have noted in the introduction section, P is an essential element/nutrient to plants, shortage of which results to deficiency symptoms. Its roles in the plant include;

- P is essential component of adenosine triphosphate (ATP) –the energy currency of the living cell. ATP contains a high-energy phosphate group that drives most energy –requiring biochemical processes e.g. the uptake of plant and their transport within the plant.
- P is an essential component of deoxyribonucleic acid (DNA), the seat of genetic inheritance and of ribonucleic acid (RNA), which directs protein synthesis in both plants and animals.
- Adequate supply enhances photosynthesis, nitrogen fixation, flowering and fruiting and maturation.
- Root growth, particularly development of lateral roots and fibrous roots is encouraged by P.
- In cereal production, good phosphorus nutrition strengthens tissues thus help prevent lodging.

Short supply of P leads to deficiency symptoms. You should know that P deficiency is generally not as easy to recognize as are deficiencies in many other nutrients. However, when you work into a field of crops and observe the symptoms below, it may certainly be P deficiency symptoms:

- Phosphorus deficient plant is usually stunted, thin-stemmed because growth is retarded.
- Foliage is bluish, purple stems, and violet foliage, especially in the older leaves.
- Yield of flowers and fruits is poor e.g. reduced tillering in cereals.
- Premature drop of fruits and flowers may occur.
- Poor seed development.



4.2 The Phosphorus Cycle

The **phosphorus cycle** is the biogeochemical that describes the movement of P through the lithosphere, hydrosphere and Biosphere. Unlike many other biogeochemical cycles, the atmosphere does not play a significant role in the movements of phosphorus, because phosphorus and phosphorus-based compounds are usually solids at the typical ranges of temperature and pressure found on Earth (Figure 4.1).

Phosphorus in the soil solution

- Many phosphate compounds are not very soluble in water; therefore, most of the phosphate in natural systems exists in solid form
- P in the soil solution is generally low
- Plant roots absorb phosphorus dissolved in the soil solution, mainly as phosphate ions (HPO_4^{2-} and H_2PO_4^-)
- In strongly acid soils (pH 4 to 5.5), the monovalent ion (H_2PO_4^-) dominates, while in alkaline solution, the divalent ion (HPO_4^{2-}) dominates.

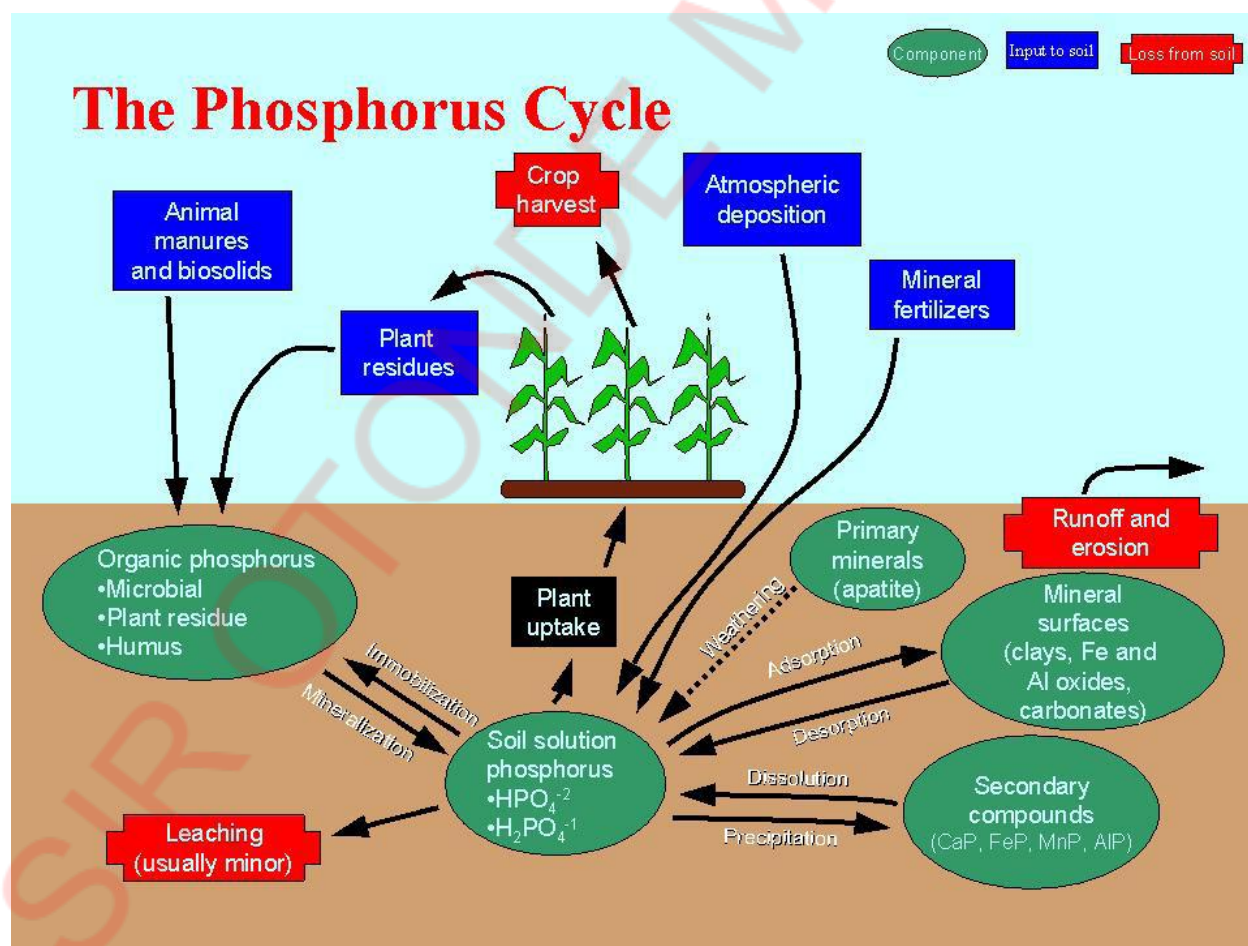


Figure 4.1: The phosphorus Cycle

Chemical forms in soil

In most soils, the amount of P in soil solution for plant uptake is low; seldom exceeding 0.01% of the total P in the soil. The bulk of soil P exists in three general groups – namely **organic phosphorus**, **calcium-bound inorganic phosphorus**, and **iron or aluminium-bound inorganic phosphorus**. Unlike nitrogen and sulfur, phosphorus is not lost from the soil in gaseous form. Because mineral surfaces strongly adsorb soluble inorganic forms of P, there is no leaching of P. The principle pathways by which P is lost from the soil system are plant removal, erosion of Phosphorus-carrying soil particles, and phosphorus dissolved in runoff

Organic P in soils

- Both inorganic and organic forms of P occur in soils, and both are important to plants as sources of this element.
- The relative amounts vary – the organic fraction generally constitutes 20 -80% of the total P in surface soil horizons.
- The deeper horizons may hold large amounts of inorganic P, especially in soil from arid and semi-arid regions
- Phosphorus held in organic form can be mineralized and immobilized

Organic P forms $\longrightarrow$ H_2PO_4^- $\longrightarrow$ Fe, Al, Ca phosphates

- Soluble P compounds (phosphate ions) are released when organic residues and humus decompose
- The phosphate ions are subject to uptake by plants or to fixation into insoluble forms by reaction with iron, aluminum, manganese and calcium in soils.
- If organic residues added are high in carbon and other nutrients, microbes increase their activity and immobilize the P in their biomass.
- The readily decomposable or easily soluble fractions of soil organic P are often the most important factor in supplying P to plants in highly weathered soils (eg Ultisols, and Oxisols)
- Mineralisation of organic phosphorus in soils is subject to temperature, moisture, tillage etc
- In temperate regions, mineralization of organic P in soils typically releases 5 to 20 kg/ha annually while in forested soils in tropical climates, the amount of P released by mineralization may exceed 60 kg /ha annually.

Inorganic phosphorus

- Two phenomena tend to control the concentration of inorganic P in the soil solution and movement in soils 1) **the solubility of phosphorus-containing minerals** and 2) **the fixation or adsorption of phosphate ions on the surface of soil particles**.
- Dissolved phosphate ions in mineral soils are subject to many types of reactions that tend to remove the ions from the soil solution and produce phosphorus-containing compounds of very low solubility referred to as P fixation called P fixation

- The P fixation has far reaching consequences for P management

4.3 The phosphorus problem in the soil

The P problem is **three fold**

- First the total phosphorus level is low, usually no more than one tenth to one fourth that of nitrogen.
- Second- the phosphorus compounds commonly found in soils are mostly unavailable for plant uptake because they are highly insoluble
- Third- When soluble sources of P such as those in fertilizers and manures are added to soils, they are fixed (changed to unavailable forms). Fixation only allows a small fraction of the P in fertilizer to be taken up by the plants

The P fixation problem

In areas of developed nations, where farmers can afford fertilizer, they apply excess P over the years, hence saturating the P fixation capacity and this helps build up levels of soil phosphorus. This long-term build-up of P also referred to as P capitalization has helped improve soil fertility. In Africa, scientists have proposed P capitalization but this has not been implemented as it requires funding

Factors affecting P fixation in soils

The phosphorus-fixation capacity of soil is positively related to certain soil properties such as calcium carbonate content, clay content, contents of iron, aluminum and manganese, and especially hydrous oxides of these

Amount of clay

Most of the compounds with which phosphorus reacts are in the finer soil fractions. Therefore phosphorus fixation is more pronounced in those soils that have higher clay contents.

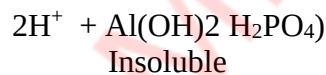
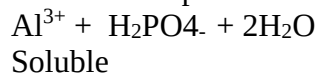
Type of clay present

- Some clay minerals are much more defective at P fixation than others.
- Clays that have greater anion exchange capacity (due to positive charges) have greater affinity for phosphate ions.
- Example is allophone clays typically found in Andisols and other soils associated with volcanic ash.
- Andisols (USDA) or Andosols (FAO) are characterized by large quantities of amorphous oxides and allophone and have the greatest Phosphorus fixing capacity, and their productivity is often limited by this property. Found in Great Rift valley, Naivasha (Kenya)

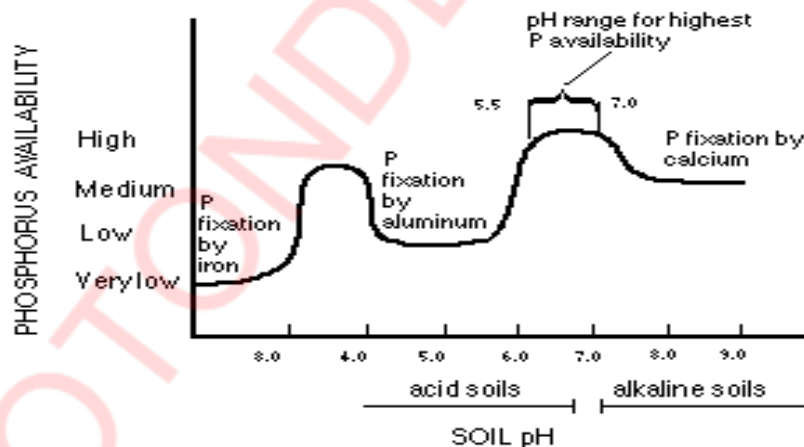
- Kaolinite, a 1:1 clay mineral mainly found in highly weathered soils (Ultisols and Oxisols) has greater fixation than 2:1 clays which are predominant in Vertisols and Mollisols
- Also, iron and aluminum oxides are prominent in Ultisols and Oxisols, hence high fixing capacity

Effect of soil pH

- The greatest degree of phosphorus fixation occurs at **very low** and **very high** soil pH
- As pH increases to around 5 to about 6.0, the iron and aluminium phosphates become somewhat more soluble.
- In mineral soils, phosphate fixation is at its lowest (and plant availability is highest) when soil pH is maintained in the 6-7 range
- In acid soils these reactions predominantly involve Al, Fe, or Mn either as dissolved ions, as oxides or as hydrous oxides
- Example:



- In alkaline soils the reaction primarily involve precipitation as various calcium – phosphate minerals
-



The availability of phosphorus is affected by soil pH.

Effect of organic matter

- Organic matter generally has little capacity to strongly fix phosphate ions.
- Soils high in organic matter, exhibit relatively low levels of phosphorus fixation

Practical control of phosphorus availability

Recap:

- Most mineral soils have a large capacity to remove P ions from solution and fix them on particle surfaces, making them unavailable
- The P in fertilizers is also subject to fixation on soil particle surfaces, and only 10 to 15% of the phosphorus added in fertilizer is likely to be taken up by plants
- Consequently, phosphorus availability is a limiting factor in many agro-ecosystems.
- A number of approaches have been suggested to ameliorate phosphorus fertility problem

Approaches

1. **Saturation of phosphorus fixing capacity (P capitalization).** This is achieved by adding enough phosphorus to satisfy the fixation sites. This can be achieved by adding all at once with one or two massive doses of Phosphorus (usually as P fertilizer, phosphate rock or high-phosphorus animal manure) over a period of years. The slow build up will maintain ample amounts of available phosphorus because the fixation sites will have been occupied. This approach is expensive, requiring initial capital for P fertilizer, for this reason, it is not practical for most of the world's Farmers but has been put into practice in some intensive agricultural systems. A potential problem is water pollution
2. **Placement of P fertilizer.** Proper placement should minimize opportunity for fixation. Fertilizer placed in a localized rooting zone instead of mixing with the soil (broadcast) reduces contact with soil, hence minimize fixation. The recommended procedure for P fertilizer application is band application
3. **Combination of ammonium and phosphorus fertilizers:** Using ammonium in the same band with phosphorus fertilizers greatly increases root uptake of P, especially in alkaline soils. The increased uptake is probably related to the acids that roots produce when they take up N. Diammonium phosphates offer this advantage
4. **Choose phosphorus efficient plants:** some plants are able to thrive in soils with low phosphorus for example those dependent on mycorrhizal associations. Others produce root secretions that solubilize the phosphate compounds eg pigeon pea
5. **Increased cycling of organic phosphorus:** mineralization of organic materials added to soils may release phosphorus gradually in synchrony with plant needs. This slow release allows plants to take up P before it is fixed. Return of crop residues, inclusion of green manure, crops in rotations, mulching and adding animal manures and other organic wastes can all increase available phosphorus
6. **Control of soil pH:** Some control over phosphorus availability can be obtained by maintaining soil pH between 6.0 and 7.0. Maintaining of pH in this range is generally more practical in less weathered soils than in highly weathered soils of warm, humid regions. Proper liming can contribute to improved P availability in many cases.
7. **Enhancement of mycorrhizal associations;** Mycorrhizae can greatly enhance the ability of plants to obtain P from soils with low levels of Phosphorus. In the case of many shrubs and trees, inoculation of seedlings with efficient strains of mycorrhizal fungi is a practical measure

In summary, maintaining sufficient supply in a soil is basically two fold 1) addition of phosphorus-containing fertilizers or amendments and 2) regulation of soil processes that fix both added and native P.

CHAPTER 5: ACIDIC SOILS

Introduction

In this lesson we shall look at the acid soils in reference to soil fertility. The purpose of this lesson is to enable you to understand how the acid and salt-affected soils can be managed to enhance soil fertility.

Expected Learning Outcomes

By the end of this lesson, you will be able to:

1. Define soil acidity and describe its significance in soil fertility and plant growth.
2. Explain the main processes and factors that lead to changes in soil pH over time.
3. Discuss the problems associated with acid-soil infertility, including nutrient deficiencies, toxicities, and biological constraints.
4. Describe the management practices and amendments used to overcome acid-soil infertility and improve crop productivity

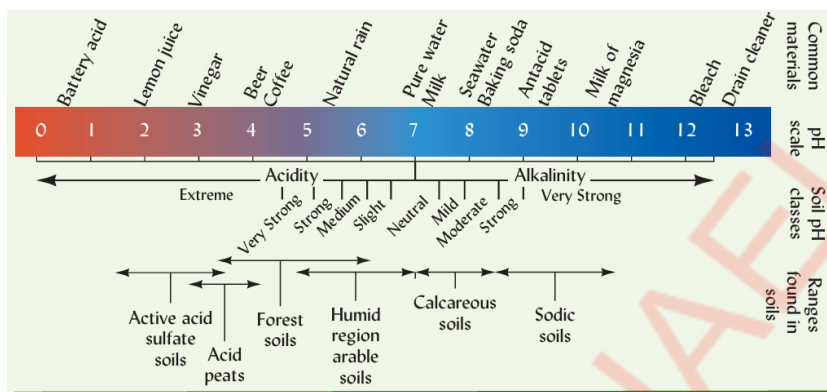
5.1 What is soil acidity

Soil acidity refers to a condition in which the soil has a **pH value below 7.0**, meaning it is more acidic than neutral. This acidity is largely due to a **high concentration of hydrogen ions (H^+)** in the soil solution. The more hydrogen ions present, the lower the pH and the more acidic the soil becomes

Soil pH is a critical factor influencing a variety of soil properties, including chemical, biological, and microbial processes. It determines plant nutrient uptake, impacts soil microorganisms, and affects plant species distribution.

Soil pH is largely controlled by balances between acid and non-acid cations on colloid surfaces and H^+ and OH^- ions in soil solutions. Acidification is most pronounced in humid regions due to leaching, whereas drier regions tend to have alkaline soils. Understanding these dynamics is essential for managing soil acidity and crop production.

Figure 9.2 Some pH values for familiar substances (top) compared to terms used to describe soil acidity or alkalinity and ranges of pH typical for various types of soils (bottom).



Soil pH levels

A pH range between 5 and 6 is considered ideal for most plants. Acid soils have a major effect on plant productivity once the soil pH falls below 5:

- pH 6.5 — **close to neutral** — Optimum for many acid-sensitive plants. Some trace elements may become unavailable.
- pH 5.5 — **slightly acid** — Optimal balance of major nutrients and trace elements available for plant uptake.
- pH 5.0 — **moderately acid** — Below pH 4.8, aluminium (Al) can become toxic to plants, depending on soil type. Phosphorus combines with Al and may be less available to plants.
- pH 4.5 — **strongly acid** — Aluminium becomes soluble in toxic quantities. Manganese (Mn) becomes soluble and toxic to plants in some soils, depending on temperature and moisture conditions. Molybdenum (Mo) is less available. Soil bacterial activity is slowed down.
- pH 4.0 — **extremely acidic** — Irreversible soil structural breakdown can occur.

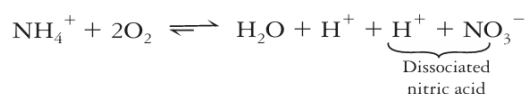
Soil pH will influence both the availability of soil nutrients to plants and the interactions among nutrients.

5.2 Causes of soil acidity

Soil acidity is caused by various activities and factors that lead to an increase in hydrogen ions (H^+) in the soil. These sources of hydrogen ions include natural processes and human activities.

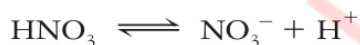
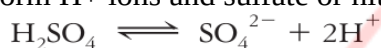
Acid-forming factors include:

1. **Rainfall:** Rainfall is one of the major causes of soil acidity mainly through leaching and acid deposition. In areas with heavy rainfall, essential nutrients and basic cations like calcium, magnesium, potassium, and sodium are washed away from the soil. This removal reduces the concentration of these basic cations, causing an increase in hydrogen ions (H^+) and resulting in lower soil pH and increased acidity.
2. **Decaying organic matter:** The carbon dioxide (CO_2) produced by decaying organic matter reacts with water in the soil to form a weak acid called **carbonic acid**. Several organic acids are also produced by decaying organic matter, but they are also weak acids.

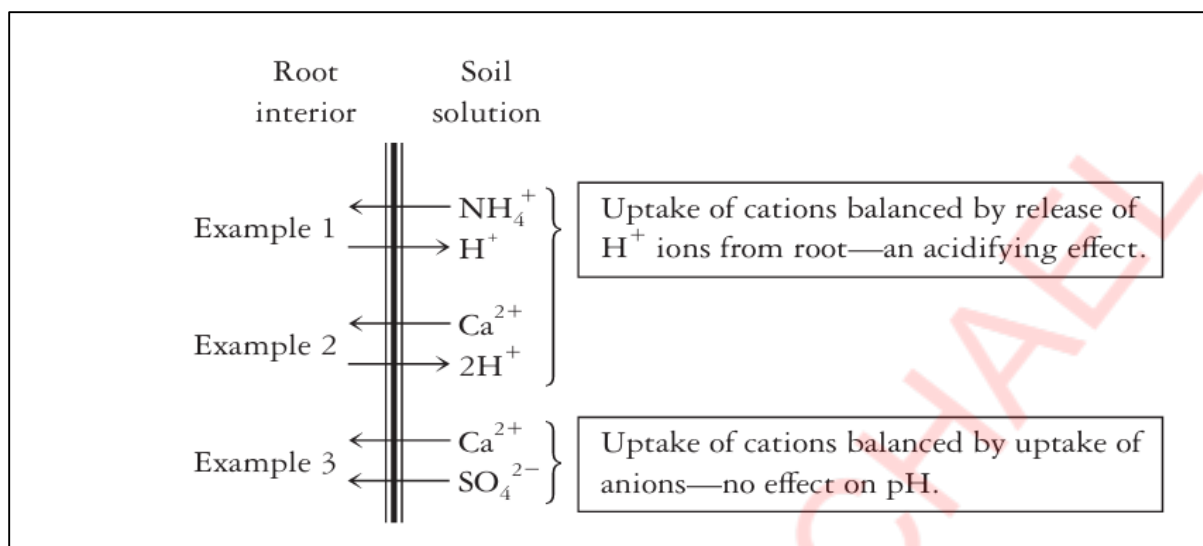


3.

4. **Carbonic and other organic Acids.** Perhaps the most widespread contributor to soil acidity is the formation of carbonic acid, which dissociates into H^+ ions. This weak acid is formed when carbon dioxide gas dissolves in water. **Root respiration** and the decomposition of soil organic matter by microorganisms produce high levels of CO_2 in soil air. In addition to carbonic acid, the breakdown of soil organic matter produces other weak organic acids, such as: humic and fulvic acids, citric acid, acetic acid, oxalic acid. These acids also release **H^+ ions** when they dissociate, adding to soil acidity.
5. **Crops removal and harvesting:** When these crops are harvested and the yield is removed from the field, some of the basic material responsible for counteracting the acidity developed by other processes is lost, and the net effect is increased soil acidity.
6. **Use of fertilizers:** Some fertilizers, especially those supplying ammonium and urea, are associated with soil acidity. Acidity is produced when ammonium-containing materials are transformed into nitrate in the soil. The more ammoniac nitrogen fertilizer is applied, the more acidic the soil gets.
7. **Oxidation of Sulfur.** The decomposition of plant residues commonly involves the oxidation of organic-SH groups to yield sulfuric acid (H_2SO_4). Another important source of this strong acid is the oxidation of reduced sulfur in minerals such as pyrite
8. **Acids in precipitation.** Rain, snow, fog, and dust contain a variety of acids that contribute H^+ ions to the soil receiving the precipitation. As raindrops fall through unpolluted air, they dissolve gases (CO_2 , S, NO_2) and form enough carbonic acid, sulfuric and nitric acids. Unlike carbonic acid, these strong acids completely dissociate to form H^+ ions and sulfate or nitrate anions:



9. **Plant uptake of cations.** Plant roots affect pH. For every positive charge taken in as a cation, a root can maintain charge balance either by taking up a negative charge as an anion or by exuding a positive charge as a different cation. If the plant absorbs **more positive charges than negative charges**, it cannot stay electrically balanced internally. To compensate, it **exudes H^+ ions into the soil solution**.



5.3 Acid Soil infertility

Acid soil infertility refers to the reduced ability of acidic soils (typically with $\text{pH} < 5.5$) to support healthy plant growth

Acid soils become infertile not because they lack nutrients entirely, but because **nutrients become unavailable, toxic elements increase, and soil biology is disrupted.**

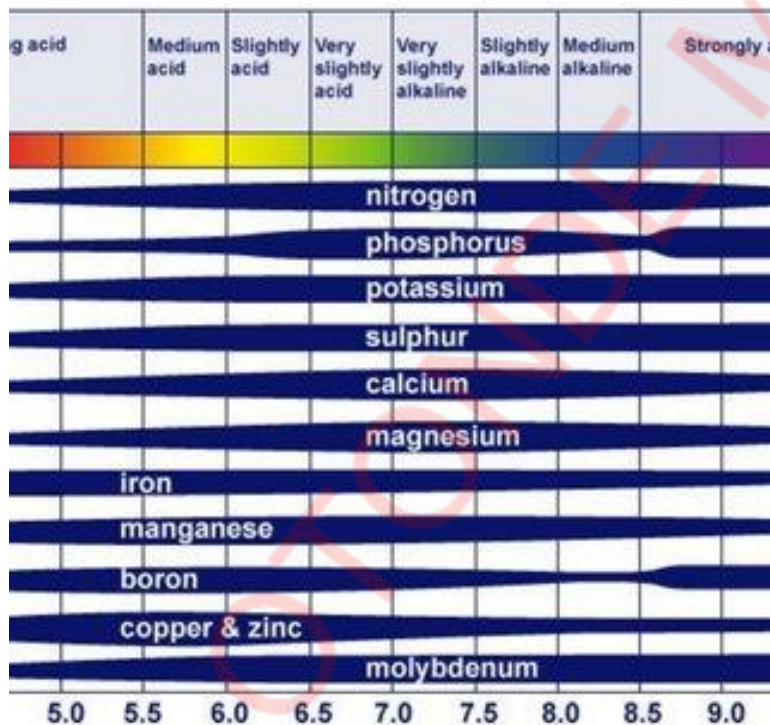
Acid soil infertility is caused by: Al toxicity, Ca & Mg deficiencies, & Mn toxicity. **Al toxicity** leads to declined yields through direct injury on roots (tobacco), resulting in restricted/retarded root development, roots becoming thicker, shorter, and stubbier, and showing dead spots.

Al accumulates in the roots, impeding the uptake & translocation of Ca & P to the leaves, and may cause their deficiency in plants. Soluble Al also reacts with P reducing its availability. Below pH 5, Al^{3+} is the dominant form of Al in soil solution. Forms of Al in solution when pH is above 5 are less toxic. **Mn toxicity** increases with decreases in soil pH; its toxicity can occur along with Al toxicity at pH lower than 5.5. In strongly acid ultisols, plant growth is commonly limited by toxic levels of Al &/or Mn before Ca &/or Mg deficiencies occur. **Molybdenum (Mo)** anions are strongly adsorbed on pH-dependent clays and Mo tends to be less available and more deficient in acid soils. Availability of Mo important for legumes because of the role it plays in N_2 -fixation (most rhizobia strains are acid-sensitive).

Few plants grow well outside the range of pH 5-8 e.g., legumes particularly sensitive to low pH and bacterial activity also suffer at low pH. Soils acidify naturally under vegetation and the process is accelerated in production agriculture. Soil pH has direct effect on plant roots; in alkaline soils, high levels of Na^+ cause toxicities while in low pH Al^{3+} reaches toxic levels near 10 μm . Mineralization & nitrification also contribute to soil acidification. NO_3^- , a strong acid is subject to leaching and is accompanied by cations (mainly K^+); H^+ accumulates in the soil

In summary soil affected by acid soil infertility express the following of characteristics

1. Aluminium toxicity. Affects plant roots Sparsely branched large diameter roots at low pH.
2. Tissue levels of Al relate poor to toxicity.
3. Manganese toxicity. ;Affects plant tops. Occurs below pH 5.5. Cotton, soybean, and tobacco are sensitive.
4. Iron toxicity but minor problem:
5. Calcium and magnesium deficiency. Peanut, tobacco, and tomato are sensitive.
6. Molybdenum deficiency (soybeans and cauliflower).In most cases liming to pH 5.5 to 6.0 alleviates Mo deficiency. In some cases liming > 6.0 needed to eliminate Mo deficiency.
7. Nitrogen, phosphorus, and sulfur deficiency because of slow organic matter decomposition is a minor problem.
8. Reduced phosphorous availability due to precipitation with aluminum.
9. Poor nodulation of legumes.



Ways of overcoming acid soil infertility

Soil acid infertility can be overcome in several ways including:

- Liming** (increases soil pH) – This is done to neutralize excess Al ions.

$$\text{CaCO}_3 + \text{H}_2\text{O} \rightarrow \text{Ca}^{2+} + \text{HCO}_3^- + \text{OH}^- (\text{OH}^- \uparrow \text{pH})$$

CaCO_3 (lime) takes several months to react completely in the soil and depends on; (a) how fine it is, - the finer it is the more rapidly it will react with the soil and (b) how well it was incorporated. Ca^{2+} displaces Al from the exchange sites. Al forms gibbsite

[Al(OH)₃], which is precipitated out & CO₂ is evolved. The desired pH rise is 5.5; if Mn toxicity is suspected, then the pH should be raised to 6. As the soluble calcium and magnesium compounds are removed from the soil by growing crops, erosion or by leaching, the percentage base saturation and pH are gradually reduced and eventually another application of lime is necessary

- ii. **Planting of tolerant crops** - roots uninjured and uptake of other nutrients not impeded eg soybean, rice, forages
- iii. **Addition of Organic Matter (OM) (high in base-forming cations)** - organic acids complex with Al reducing its toxicity. Bases are produced from decomposition and they compete for the exchange sites with aluminium. Organic matter enhance soil structure, enhance nutrient availability, and microbial activity,
- iv. **Proper fertilization:** Using fertilizers with lower acidity levels and applying them judiciously can help manage soil pH. Avoiding ammonium-based fertilizers, which can contribute to soil acidity, may also be beneficial.

Soil pH buffering

Soils have the ability to resist changes in pH when acids or bases are added, a phenomenon known as buffering. This is important because it helps maintain soil stability, ensuring that pH fluctuations do not occur rapidly, which could otherwise harm plant growth and soil microorganisms. Buffering capacity enables soils to absorb acids or bases without significant shifts in pH, protecting plants from nutrient imbalances and toxic elements. It also helps manage agricultural inputs like fertilizers and lime, allowing for more effective pH control over time. Additionally, buffering plays a key role in mitigating environmental impacts, such as acid rain or contamination, by regulating soil pH and preventing drastic changes.

Example: In some areas of Kenya, even though a soil pH of 5.1 is considered acidic, maize can still grow normally when the soil has a **good buffering capacity**. This is because the soil's clay particles, organic matter, and exchangeable nutrients help prevent sudden changes in pH around the maize roots. As a result, harmful conditions such as aluminum toxicity are minimized, and essential nutrients remain available. The buffering action keeps the root environment stable enough for maize to grow and produce reasonable yields despite the moderately acidic pH.

CHAPTER 6: SOIL FERTILITY ASSESSMENT

Learning outcomes

By the end of this chapter, students should be able to:

1. Define soil fertility assessment and explain its importance.
2. Explain the stages of soil fertility assessment from sampling to analysis.
3. Identify types of soil sampling and select appropriate methods.
4. Outline soil sample processing and analysis procedures.
5. Explain how to determine soil pH, nitrogen and phosphorus.

Importance of soil fertility assessment :

- Determines the nutrient status of the soil.
- Identifies limiting nutrients that may restrict crop growth.
- Guides fertilizer and soil amendment recommendations.
- Supports sustainable soil management and improved crop productivity.

Always remember that :

- Fertility assessment is crop-specific; different crops have different nutrient requirements.
- Includes evaluation of soil chemical, physical, and biological properties.
- Helps in efficient and economical use of fertilizers.
- Can prevent over-fertilization or nutrient depletion, protecting the environment.

6.1 Methods of evaluating soil Fertility Status

Several methods are used to indicate the fertility status of the soil. They are based on soil physical characteristics; crop performance, and presence of plants species that are associated with different soil fertility levels.

Method 1: Soil physical characteristics; fertile soils are darker in colour, while less fertile soils are lighter in colour. This is an indicator of the organic matter content

Method 2: Crop performance: The performance of the crops grown on the farm is an indicator of the fertility status of the soil. Crops grown on fertile soils perform better than crops grown on infertile soils

Method 3: Using indicator plant species: Certain plant species have been established to thrive best on infertile soils. On fertile soils, these are easily out-competed by faster-growing plant species. Their nutrient requirements are also low. These include poverty grass. Plant species that have high nutrient requirements are normally found in soils that can sustain the required nutrient supply.

Method 4: Observation of Nutrient Deficiency Symptoms

Growing plants display observable signs and symptoms in response to the unavailability of nutrients in the soil. Experience in observation is important in identifying deficient nutrients in crops. The dominance of a particular plant may also be indicative of the fertility status e.g. poverty grass. As a solution to the farmers' soil nutrient status, this method provides the least cost option. It has however several disadvantages such as;

- Late appearance of symptoms that may delay taking corrective measures. Even if taken, the particular crop may not enjoy the benefit
- Several nutrient deficiency symptoms are similar in some crops. One is unsure which nutrient to add. However, when the nutrients are added, one may be incorrectly added
- Some deficiency symptoms resemble pests/disease symptoms
- Soil fertility status cannot be derived from this method (determination of amounts of nutrients inherent and the amounts present)

Method 5: Plant Tissue Analysis

Plant analysis is a technique used to determine the nutrient status of plants by analyzing their tissue composition. This involves collecting plant tissue samples and analyzing them for various nutrient elements such as nitrogen (N), phosphorus (P), potassium (K), calcium (Ca), magnesium (Mg), and other micronutrients like iron (Fe), manganese (Mn), zinc (Zn), copper (Cu), and boron (B).

- Plants take up the nutrients required from the soil via the roots. By analysing the amount of nutrients in the leaves (as plant tissue) one can determine the nutrient levels in the soil.
- The method is based on the principle that the amount of nutrients extracted by chemical reagents from the plant tissue is equal to the amount available and supplied by the soil. Recall that deficiency symptoms are observed in the plant tissue, being also evidence of insufficient availability and supply from the soil. The method is crop and nutrient specific, and very precise for the crop in question.

Plant analysis can be done at different stages of plant growth and development, including at planting, mid-season, and post-harvest. The timing of plant analysis depends on the crop and the specific nutrient being analyzed. For example, nitrogen analysis is typically done at mid-season when the crop is actively growing and has reached a stage of rapid uptake.

Advantages

1. The method is precise to the specific crop therefore very good for specialized farming of high-value horticultural crops
2. Method is very good for deep-rooted crops where soil sampling and analysis would require digging profile (deep) pits that are cumbersome
3. It can be used at any growth stage of the crop, thereby providing a useful method to monitor crop performance throughout the growth period

Disadvantages

1. Dependence on type and growth stage of the plant is a major limitation; results for one crop cannot be translated to represent any other crop
2. Some plants display luxury consumption of certain nutrients. This may lead to wrong deductions on specific nutrients status of the soil in question

3. Though the results are nutrient specific and precise, it is costly for small-scale farmers. It is most suitable for specialized crop farming such as flower farming

Method 6: soil nutrient analysis

Soil analysis or testing is critical to soil fertility evaluation and nutrient management in agriculture. It involves **analyzing soil samples to determine the levels of various soil nutrients**. Soil testing is essential to ensure that crops receive the proper amount of nutrients for optimal growth and development while minimizing environmental impacts such as nutrient runoff and groundwater contamination.

There are various methods for soil testing, including **chemical extraction, biological assays, and spectroscopy**. Chemical extraction is the most common method used in soil testing, which involves mixing soil samples with specific chemical solutions to extract soil nutrients. The extracted solutions are then analyzed for nutrient concentrations using various analytical techniques, such as colorimetry, titration, or spectrophotometry. These methods allow for precise quantification of soil nutrients, making it easier to develop **nutrient management plans** based on crop requirements.

Soil testing should be conducted regularly, ideally every one to three years, depending on the crop type, soil type, and management practices. Soil samples should be collected from representative areas of the field, and multiple samples should be taken and combined to obtain a **representative sample**. Soil testing laboratories can guide how to collect and prepare soil samples for testing.

The results of soil testing provide valuable information on the nutrient status of the soil, which can be **used to develop nutrient management plans**.

Interpretation of analytical results and fertilizer recommendations

The results are interpreted and **recommendations** made. The nearest agricultural officer (from extension or research) can be consulted for the interpretation. This is a very crucial step so that the corrective measures are correctly undertaken to enhance soil productivity.

Benefits of soil testing

- It reveals the **cause** of nutrient problems
- The farmer will apply the right type and amount of fertilizer (no wastage due to over or under application or wrong type of fertilizer application)
- The application of the recommended fertilizer will be at the right time
- The farmer is advised on the other soil amendments/reclamation needs, e.g., liming
- Farmers advised on good soil management practices
- Healthy and good quality crop realized due to balanced nutrition
- Increased yields realized
- Intensive utilization on the static land resource as the population increases

6.2 Stages of Soil nutrients analysis

Steps in soil nutrient analysis

1. **Sample the soil:** Collect a representative sample.
2. **Mix and prepare** the composite sample, package and label the sample
3. **Deliver to Laboratory;** Deliver the sample to a reliable soil testing laboratory along with the required information sheet.
4. **Prepare soil for analysis:** air-dry, grind and sieve the sample
5. **Analyse:** The lab analyzes soil for pH, organic matter, available nutrients (N, P, K, etc.), and other properties.
6. **Interpretation of Results:** test results are compared with standard critical levels to determine nutrient status.
7. **Recommendation:** Provide fertilizer and lime recommendations based on crop requirements and soil test results.

Soil Sampling

Soil sampling is a systematic collection of soil samples in a farm for analysis. **Soil analysis** is the chemical, physical or microbiological technique that estimates the availability of essential nutrients and organisms in the soil for plant growth and predicting the nutrient requirement of crops.

Tools for soil sampling

In general tools and accessories for soil sampling depends on the purpose and conditions of the soil. The most commonly used tools and accessories are:

- a) Soil auger (spade or panga)
- b) Trowel/strong rod for mixing soil
- c) A clean bucket or any clean container
- d) Markers for labelling
- e) Soil sampling information sheet
- f) Soil sampling bags/Clean plastic containers
- g) Cooler box
- h) Core rings
- i) Tape measure/ruler
- j) Knife/sample divider
- k) Data sheet/note book
- l) GPS m) Scoopers
- n) Sampling probe/core.

Preparation Phase:

- Define objective (routine analysis or research).
- Plan sampling locations and depth.
- Clean and prepare sampling tools.
- Prepare a short sampling protocol.

Field Sampling phase:

Types of Soil Sampling (Four main types)

1. Grid Sampling: Field divided into uniform grids; sample taken from each grid. Each sample is a composite of 6–8 cores (or points in a circle). Can use GPS for precise mapping of field characteristics.
2. Zone Sampling (by landscape/topography): Field divided into zones based on topography, cropping history, or management. Examples: uplands, slopes, bottomlands. Nutrient levels vary by zone: highest in bottomlands, moderate in flat areas, lowest in top/eroded areas.
3. Whole-Farm Sampling: Single composite sample represents the entire field (≤ 25 acres). Assumes uniform soil conditions across the field.
4. Experimental plot sampling: This type of sampling is used in research experiments where the field is divided into **clearly defined plots** with different treatments (e.g., fertilizer rates, crop varieties, tillage methods). Each **experimental plot** is sampled **separately** to evaluate treatment effects. Samples are taken systematically within each plot, usually using **5–10 cores** to form a composite sample per plot. Plots are typically small and managed uniformly within each treatment.

70% of the errors in soil test results and recommendations arise from this stage. One must be careful how it's done.

Errors can easily be minimized if the following areas are avoided:

- Livestock bomas, (current and past)
- Charcoal burning sites
- Homesteads
- Livestock movement paths in grazing areas
- Manure heaps
- Old fertilizer bands
- Field borders especially if close to gravel or road
- Animal excretion spots
- Bush/trash/straw/hay piles
- Trouble spots such as ant hills, low spots, salinity spots, eroded areas, or any other unusual areas unless separately sampled

Key steps to sampling

1. Choose method: grid, whole-farm, or zoning.
2. Divide field into homogeneous units.

3. Remove surface litter.
4. Collect soil with auger or 'V'-shaped spade cut.
5. Take 10–15 subsamples per unit; mix thoroughly.
6. Remove debris (roots, stones, pebbles).
7. Reduce to 1kg to ~0.5 kg via quartering; store in clean bag.
8. Repeat in other field areas.
9. Label samples with farmer, location, crops, date, sampler.
10. Transport to laboratory for analysis.

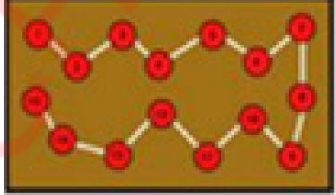
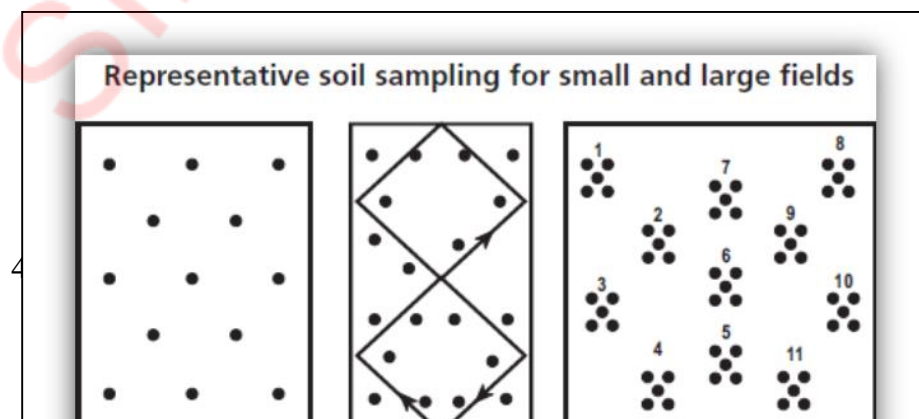
1 	2  or Sample to a depth of 0-20 cm for shallow and > 20 cm for deep rooted crops	3  Take soil from at least 10 places in each field
Label the bag: • Name of the farmer • Sampling date • Location/GPS • Sampling depth • Analysis required • Previous crop • Crop to be planted Insert a similar tag in the sampling bag	5  Put about 500 g (1/2 kg) of the composite soil in a sampling bag	6  Mix the soil thoroughly to get a composite sample

Figure shown provides the suggested sampling procedure for a small field (Finck, 1992) and for a large field (Peck and Melsted, 1967).



Soil Sample Processing in the lab

- 1. Sample Preparation:** Remove debris and air-dry the soil at room temperature; avoid direct sunlight or high heat. Then crush soil clods gently and sieve (usually 2 mm) to obtain uniform particle size. Mix thoroughly to ensure homogeneity.
- 2. Sample Subdivision:** Reduce sample size to the amount needed for analysis (usually ~0.5 kg). Store in clean, labeled containers (cloth, paper, or plastic bags). Proper labeling: farmer, location, field, crop history, date of collection.
- 3. Storage:** Keep samples dry and protected from contamination.

Soil analysis methods

- **Purpose:** To measure soil properties that affect fertility.
- **Chemical Analysis:**
 - pH (soil acidity/alkalinity), electrical conductivity (salinity).
 - Macronutrients: Nitrogen (N), Phosphorus (P), Potassium (K).
 - Secondary nutrients: Calcium (Ca), Magnesium (Mg), Sulfur (S).
 - Micronutrients: Iron (Fe), Zinc (Zn), Manganese (Mn), Boron (B), Copper (Cu), Molybdenum (Mo).
- **Physical Analysis:**
 - Soil texture (sand, silt, clay proportion).
 - Bulk density and porosity (affect water and nutrient retention).
- **Biological Analysis:**
 - Organic matter content (affects nutrient supply and soil structure).

Determination of soil pH

Soil pH is a measure of the acidity or alkalinity of a soil. Chemistry laboratories generally measure soil pH using both water and calcium chloride. The simplest method is to measure pH_w with a portable pH meter. The soil sample is prepared with either water or dilute salt solution, as the electrode only measures acidity in solution rather than H⁺ ions held on the soil surface (Okalebo et al., 2009)

Determination of total Nitrogen

Nitrogen analyses are not difficult to conduct, but interpreting results can be problematic. This is because plant availability of soil N depends on OM breakdown, which can not be predicted from a soil test. Nitrogen in the nitrate form (NO₃-N) is directly available to plants, however, NO₃-N can be quickly lost from soil. Be aware that nitrate analyses provides a 'snapshot' of available N, but may not indicate N availability later in the growing season.

Common methods used to measure total N in soils include:

1. Kjeldahl digestions (Total Kjeldahl Nitrogen – TKN)

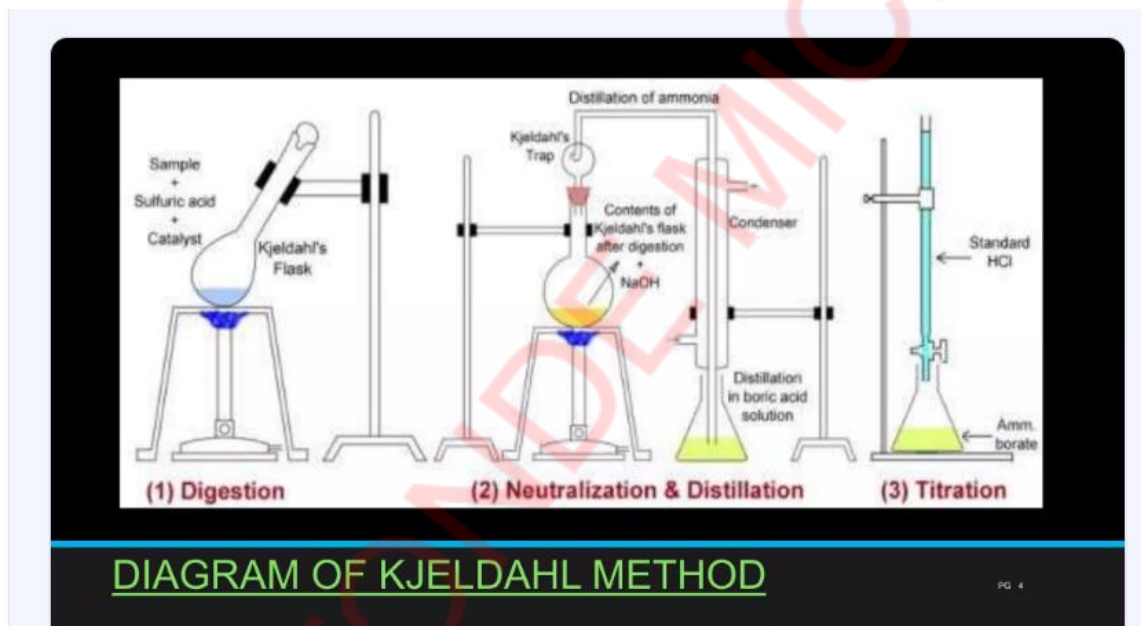
2. Carbon-Nitrogen analyzer (CN Analyzer)

Kjeldahl Digestion (Total Kjeldahl Nitrogen – TKN):

1. **Digestion:** Soil/plant sample + H_2SO_4 + catalyst $\rightarrow$ converts organic N to ammonium sulfate.
2. **Neutralization & Distillation:** Add NaOH to convert ammonium to NH_3 ; distill into boric acid solution.
3. **Titration:** Titrate trapped NH_3 with standard HCl or H_2SO_4 ; calculate % N.

Advantages:

- Measures **total nitrogen** (organic + ammonium).
- Reliable and widely used in soil and plant analysis.



Determination of Phosphorus (P)

Most soil P is tightly bound to soil particles. The P-containing complexes in alkaline soils are very different than those in neutral or acidic soils. The amount of P removed during soil extraction is dependent on the nature of P complexes and on the specific extractant used, so P extractants must be matched to soil properties.

Two key activities Extraction and measurement of P

Extraction of Soil P:

Choose a suitable extracting solution based on the soil type, pH, and expected P content.

Common extracting solutions include Bray P1, Bray P2, Olsen P, and Mehlich-3. Steps include

1. Weigh a known amount of soil (e.g., 5 grams) into a beaker or extraction vessel.
2. Add the extracting solution to the soil sample in the appropriate ratio, typically based on a soil-to-extractant ratio (e.g., 1:10 or 1:20).

3. Shake or stir the soil-extractant mixture for a specified period (e.g., 30 minutes to 1 hour) to facilitate the extraction of P from the soil.
4. After extraction, centrifuge or filter the soil-extractant mixture to separate the soil particles from the liquid extract.

Measurement of Extracted P:

Measure the concentration of P in the extract using a suitable analytical method. Common methods include colorimetry or spectrophotometry. Measure the absorbance or emission intensity of the extract at the appropriate wavelength and compare it to the calibration curve to determine the concentration of P in the

Interpretation of soil analysis results

Interpretation of Soil Analysis for Fertilizer Recommendations – Summary

- After soil analysis, **nutrient concentrations** are interpreted to assess soil fertility and determine required nutrient applications for target yields.
- **Low soil test values** indicate nutrient deficiencies, requiring higher external fertilizer application to achieve optimal yields.
- **High soil test values** suggest nutrient sufficiency; little or no fertilizer is needed.
- In advanced agricultural systems, nutrients like phosphorus may still be applied to **replace crop removals** and maintain soil fertility, though this may be unaffordable for some farmers.
- **Macronutrients** are categorized into: very low, low, medium, high, very high, guiding application rates.
- **Micronutrients** rely on a **critical level** to determine the need for supplementation.
- Soil fertility categories can be linked to **expected crop yields** without external nutrient addition, providing a basis for fertilizer planning.

Interpretation of soil test data for some nutrients in soils with medium CEC

Soil fertility class	Available (extractable) nutrients			Expected relative yield without fertilizer (%)
	P	K	Mg	
	(mg/kg soil)			
Very low	< 5	< 50	< 20	< 50
Low	5–9	50–100	20–40	50–80
Medium	10–17	100–175	40–80	80–100
High	18–25	175–300	80–180	100
Very high	> 25	> 300	> 180	100

Source: FAO, 1980.

In most cases, soil nutrient status is stated as low, medium or high. This needs to be done for each nutrient. For nutrients other than N, P and K, a single critical level is usually designated below which a soil is considered to be deficient in that nutrient, hence requiring its application.

CHAPTER 7: NUTRIENT UPTAKE BY PLANTS

Lesson Learning Outcomes

By the end of this lesson, you will be able to:

1. Explain mechanisms in which ions move from soils to roots and their absorption by plants.
2. Critically discuss nutrient uptake in time and contents.
3. Explain mobility of nutrients and their deficiency symptoms.

In this, we are discussing how plants acquire nutrients from both the soil and the air/atmosphere for metabolism processes, and the creation of new and/or the replacement of worn-out tissues.

7.1 Mechanisms in which ions move from soils to roots

Plants obtain nutrients primarily through their roots. Studies have found that the movement of ions is triggered by the formation of sugars in the leaves and their transport to the phloem.

Let us appreciate the fact that the movement of nutrients in the soil basically depends on three factors, which include;

- i. their concentration in soil solution;
- ii. how strongly clays and organic matter, and adsorb them
- iii. how fast they can move.

There are three ways in which ions in the soils may reach the root surface. These are what we are referring to as the mechanisms. Other authors refer to the process as movement of ions within the soil. Regardless of how they are referred to, they are main three ways/mechanisms/methods and they include;

- i. Mass Flow
- ii. Diffusion
- iii. Root interception,

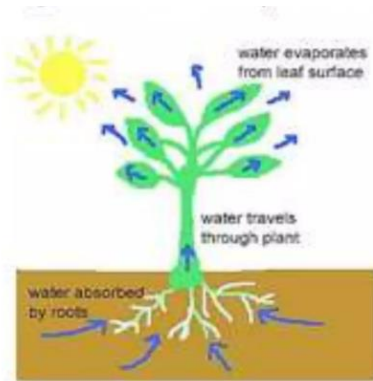
1. Mass flow

Mass flow is the bulk movement of water and dissolved nutrients toward plant roots, driven mainly by transpiration. It is crucial for delivering nitrate, calcium, magnesium, sulfate, and some micronutrients, and its efficiency depends on soil moisture, nutrient solubility, and plant water uptake

How Mass Flow Works

- When plants transpire, water is drawn from the soil into the roots and moves upward through the xylem to the leaves.
- As water flows toward the roots, it **carries with it dissolved mineral nutrients** present in the soil solution.

- These nutrients are then absorbed by root cells either passively with the water or actively via transport proteins in root membranes.



Nutrients Supplied by Mass Flow: Mass flow is especially important for **mobile and soluble nutrients** that remain dissolved in soil water. Examples include:

- Nitrate (NO_3^-)
- Calcium (Ca^{2+})
- Magnesium (Mg^{2+})
- Sulphate (SO_4^{2-})
- Some micronutrients such as **boron (B)** and **molybdate (MoO_4^{2-})**.

Less mobile nutrients such as phosphorus (P) and potassium (K) are supplied mainly by **diffusion**, not mass flow.

Factors Affecting Mass Flow

- **Transpiration rate:** Higher transpiration increases water uptake and thus mass flow of nutrients.
- **Soil moisture:** Adequate moisture ensures continuous water movement; drought slows mass flow.
- **Nutrient concentration in soil solution:** Higher solubility improves transport.
- **Soil texture and structure:** Sandy soils with good water movement favor mass flow, while compacted soils hinder it.
- **Plant growth stage:** Actively growing plants with high water demand enhance mass flow.

Agricultural Importance

- Mass flow delivers most of the **nitrogen (as nitrate)** and much of the **calcium and magnesium** needed by crops.
- It links **water management** directly with **nutrient uptake**—plants under drought stress often suffer nutrient deficiencies even if the soil is fertile.

- Efficient irrigation and maintaining adequate soil moisture are key to maximizing nutrient uptake by mass flow.

Key points to note

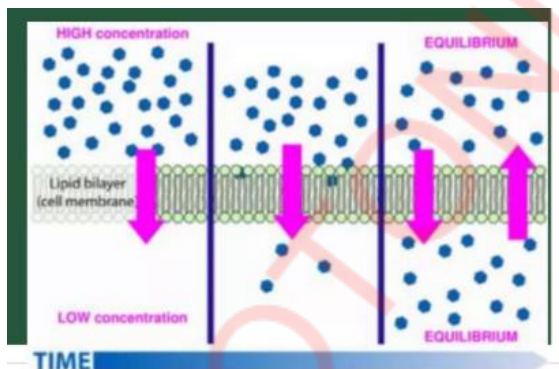
- Mass flow is driven by water uptake due to transpiration.
- It moves nutrients toward roots, not the other way around.
- Essential for supplying highly soluble, mobile nutrients.
- Strongly influenced by soil water availability and plant water use.

2. Diffusion

Diffusion involves the movement of nutrient ions from an area of higher concentration in the soil solution to an area of lower concentration near the root surface. This process happens naturally due to the concentration gradient—nutrients move until the concentration becomes balanced (equilibrium).

This is how it works

- As roots absorb nutrients, the concentration of those ions near the root surface decreases.
- The surrounding soil, farther away from the root, still has higher concentrations of these ions. Nutrient ions then diffuse through soil water toward the root hairs to replace what has been absorbed.
- Diffusion occurs even in still soil water, though it is slower than mass flow.



Diffusion is the **primary transport mechanism** for nutrients that are:

- **Less soluble** in water, or **Tightly bound** to soil particles, and therefore move slowly in the soil solution.

Key examples:

- **Phosphorus (P):** as phosphate ions (H_2PO_4^- , HPO_4^{2-})

- **Potassium (K^+)**
- **Zinc (Zn^{2+})**
- **Copper (Cu^{2+})**
- **Iron (Fe^{2+}/Fe^{3+})** in certain soils

Factors Affecting Diffusion

- **Soil Moisture:** Water forms the medium for diffusion; dry soils drastically reduce diffusion.
- **Soil Temperature:** Warm soils enhance diffusion rates, while cold soils slow the process.
- **Soil Texture and Structure:** Fine-textured soils with poor porosity can limit diffusion; good structure with adequate pores helps.
- **Soil pH and Chemistry:** Influence solubility of nutrients like P and micronutrients.
- **Nutrient Concentration:** Higher concentration gradients (from fertilizer placement) increase diffusion toward roots.
- **Root Density and Mycorrhizae:** Greater root or hyphal surface area shortens the distance nutrients need to diffuse.

Key Takeaways

- **Diffusion:** Movement of ions **along a concentration gradient** from higher to lower concentration toward the root surface.
- Critical for **less mobile nutrients**, especially **P and K**.
- Strongly influenced by **soil moisture, temperature, and nutrient concentration gradients**.
- Efficient nutrient placement and healthy root systems improve diffusion.

3. Root interception

It occurs when growing roots and root hairs physically come into contact with soil particles that contain nutrients. Unlike mass flow (driven by water movement) or diffusion (driven by concentration gradients), root interception depends entirely on root growth and exploration of new soil zones.

How Root Interception Works: As roots elongate and produce fine root hairs, they penetrate fresh areas of the soil matrix. When root surfaces touch nutrient-bearing soil particles or exchange sites (like clay minerals or organic matter), nutrient ions are absorbed directly into the roots. This process is particularly important for nutrients that are strongly adsorbed (held tightly) to soil colloids and therefore do not move easily in soil water.

Important for nutrients such as:

- **Calcium (Ca^{2+})**
- **Magnesium (Mg^{2+})**

- **Zinc (Zn^{2+})**
- **Copper (Cu^{2+})**
- **Manganese (Mn^{2+})**
- Other micronutrients that are held on cation-exchange sites.

Factors Affecting Root Interception

- Root growth rate and density: Faster-growing roots and dense root hairs increase soil contact.
- Root morphology: Fine roots and mycorrhizal hyphae increase contact with soil particles.
- Soil structure: Well-aggregated soils allow roots to penetrate easily, improving interception.
- Soil compaction: Hard or compacted soils limit root exploration and reduce interception.
- Moisture and aeration: Adequate water and air encourage deeper root growth and better soil contact.
- **Mycorrhizal fungi** extend the effective reach of roots through their hyphae.
- These hyphae grow into soil pores too small for roots, greatly increasing the soil volume explored and improving access to immobile nutrients like **phosphorus (P)** and some micronutrients.

Note

- Root interception: Nutrients are absorbed when roots physically touch nutrient-rich soil particles. It is critical to acquire fewer mobile cations and micronutrients.
- Depends on root growth, soil structure, and biological associations (like mycorrhizae).

Complements mass flow and diffusion to ensure plants get a full nutrient supply

CHAPTER 8: TYPES AND APPLICATION METHODS OF MINERAL FERTILIZER

Learning outcomes

By the end of this lesson, you will be able to:

- Identify and classify the main types of mineral fertilizers (e.g., nitrogenous, phosphatic, potassic, compound, and micronutrient fertilizers).
- Explain each fertilizer type's chemical forms, properties, and nutrient contents.
- Describe the limitations and appropriate uses of different fertilizer types for various soils and crops.
- Differentiate among the main application methods (broadcasting, band placement, side-dressing, top-dressing, fertigation, foliar feeding, etc.).

Introduction

- Mineral fertilizers are the quickest and surest way of supplying nutrients in known amounts, proportions, and forms ready for uptake by plants.
- It is acknowledged that 50% of yield increase for most crops can be attributed to efficient fertilizer use when other factors are at an optimum.
- Since inorganic fertilizers can be timely applied to coincide with crop demands at specific growth stages, they stand a chance of optimizing on fertilizer use efficiency (FUE).

8.1 Benefits of mineral fertilizers

1. Supply essential nutrients quickly for healthy crop growth.
2. Predictable nutrient composition for accurate application rates
3. Increase crop yields and improve food production.
4. Provide concentrated nutrients (small amounts give big results).
5. Correct soil nutrient deficiencies that limit plant growth.
6. Improve crop quality (better grain filling, fruit size, protein content).
7. Support continuous cropping on the same land.
8. Easy to transport, store, and apply compared to bulky organic manures..

8.2 Classification of fertilizers

Inorganic fertilizers are classified according to the following criteria:

a) Method of production

1. Naturally occurring e.g. phosphates
2. Synthetic e.g. urea, calcium ammonium nitrate

b) Number of Nutrients

Single nutrient or straight fertilizers (contain only one of N, P, or K) Examples: **Sulphate of Ammonia** ($(\text{NH}_4)_2\text{SO}_4$) → provides only N, **Muriate of Potash** (KCl) → provides only K and **Single superphosphate (SSP)** → provides only P

- Incomplete fertilizers (contains two of NPKs) e.g. DAP, 23.23.0.
- Complete fertilizer (contains all 3 NPKs) e.g. 17.17.17., 20.10.10
- Multi-nutrient fertilizer (Multiple nutrients). Contains many micronutrients e.g. Bayfolan foliar feed

c) Mode of Action

- Quick acting (water soluble and immediately available) e.g. MOP, Urea, foliar feeds
- Slow acting (transformation into soluble forms required) e.g. rock phosphate

Here we use number of nutrients

8.2.1 Nitrogen fertilizers

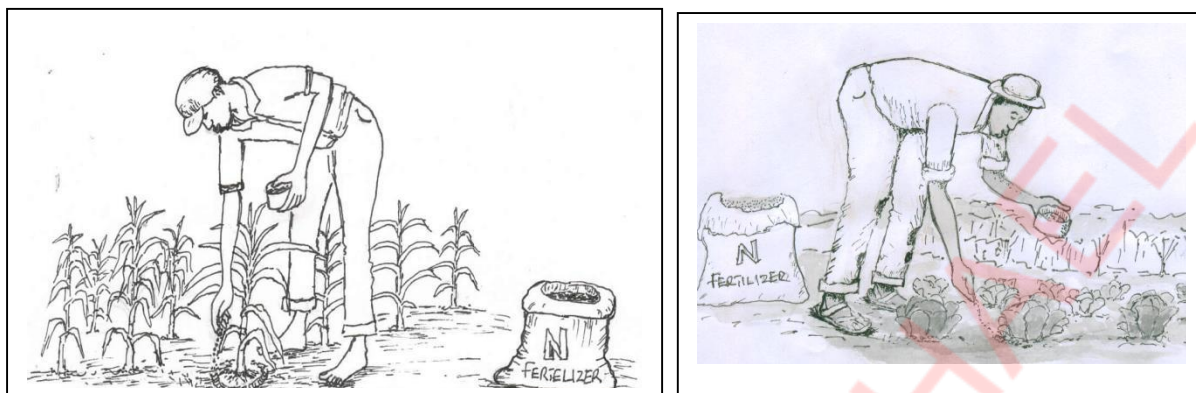
- Nitrogen fertilizers are developed to supply nitrogen (N) as the major nutrient.
- N is mobile in plants and is moved from older to young leaves during growth making deficiency symptoms to begin on leaves at the bottom of plants.
- N fertilizers are valued according to their total N content, the N forms and side effects such as acidification of the soil.
- Nitrate (NO_3^-) N is quick acting because it is immediately available.
- Plants take up N mainly in NO_3^- form but it is most leachable.
- Ammonium (NH_4^+) N has slower effect because it has to be nitrified first before N release in many cases.
- Amide N such as urea must be transformed into ammonium and finally nitrate form by soil microbial activity before plants can take it up.

Nitrogen use efficiency can be enhanced through: -

1. Matching crop need with nutrient availability
2. Application of the right type of N-fertilizer for specific crops and soils
3. Correct rates of application that are zone and crop specific
4. Correct mode and method of application for particular tillage systems
5. Proper timing of fertilizer application for nutrient exploitation
6. Matching rates of application to the target yields; avoid under or over application
7. Availability of other nutrients in adequate amounts and balanced proportions
8. Proper control of soil acidity

Mode of application

- Most N-fertilizers are applied as top-dressing when plants are actively growing and soils are moist. There are different types of N-fertilizers and a few are discussed in the following sections;



Applying top dressing N fertilizer

Examples of N fertilizers

1. Ammonium fertilizers

- Ammonium fertilizers contain nitrogen in form of NH_4^+ .
- They are moderately quick acting. Soil colloids absorb NH_4^+ , which becomes less liable to leaching.
- It is rapidly converted to nitrates (NO_3^-) by soil nitrifying bacteria.
- Some plants can take up NH_4^+ directly especially during early growth stages.
- Hence ammonium fertilizers can supply N either in NH_4^+ form or through nitrification as NO_3^- ions.
- They are acid forming and continuous applications without liming may cause soil acidity problems.
- Common ones are **sulphate of ammonia (SA)** and **ammonium chloride**. SA is an important source of N in Sulphur-deficient soils

2. Ammonium nitrate fertilizers

Contain N in both NO_3^- and NH_4^+ . Common ones are ammonium nitrate (AN), ammonium sulphate nitrate (ASN) and calcium ammonium nitrate (CAN).

Calcium Ammonium Nitrate (CAN) 26-0-0

- Half of the N is in NO_3^- and half in NH_4^+ form
- Colour: Grey or light brown depending on coat used to prevent caking
- Its free flowing but proper storage is in air-conditioned silos
- CAN is about neutral when applied to soil making it suitable even in acidic soils
- It is like ammonium nitrate except for the secondary nutrient (calcium)

Ammonium Sulphate Nitrate (ASN) 26-0-0+15S

- At least $\frac{1}{4}$ is in the NO_3^- form and about $\frac{3}{4}$ in NH_4^+ form
- Solubility: Water soluble leaving no residues

- A good source of N and S
- ASN may cake in storage and must be broken up before application
- It is applied as top dressing by broadcasting, banding or side dressing
- Intermediate leaching relative to ammonium nitrate fertilizers

8.2.2 Phosphatic Fertilizers

- The total P content in fertilizer does not guarantee its availability for crop uptake.
- The best measure of P availability is the extent of its being absorbed by plants under favourable conditions of crop growth.
- Under farming conditions on well-supplied soils, common P-fertilizers give about an equal yield response per unit of available P_2O_5 .
- However, water-soluble P is superior for annual crops and crops which have limited root systems in P-deficient soils.
- The utilization of P is only 16-22% in the first year of application and only 1-2% in each of subsequent years such that only $\frac{2}{3}$ is taken up by the end of thirty years.

Types Phosphatic fertilizers

1. Rock Phosphate

- Naturally occurring as appetites, e.g., minjingu rock phosphate
- Finely ground for direct application
- A raw material for all phosphatic fertilizers
- Its composition and suitability for direct application depends on source
- Neutral in its reaction in soil
- Has a total P content of 18-40% P_2O_5
- The mineral contains 35-38% calcium but of no liming value
- Rock phosphate is 99% insoluble in water

Application of rock phosphate

- There is little correlation between chemical analysis and P-availability to crops.
- Rock phosphate is a slow acting P-fertilizer and its efficiency depends upon:
- Rock phosphate should be broadcast and thoroughly mixed with soil for better agronomic performance.
- Dustiness makes spreading of material difficult. It must be applied 2-3 times the rate of ordinary super-phosphate for good response.
- It has long residual effect.
- It should be applied before planting to allow time for solubilisation to take place

2. Single super phosphate (SSP) 0-(16-22)-0+16S

- Product of reacting rock phosphate with sulphuric acid
- SSP contains 16-22% P_2O_5 and 16% S
- Solubility: SSP is 90% water-soluble

- SSP is less cost-effective due to its low fertilizer grade

Mode of application

- Should be applied in bands at planting close to seed rows.
- Banding minimizes the extent of soil fertilizer contact thus slowing the rate of P fixation and immobilization

3. Triple super-phosphate (TSP) 0-46-0

- Made by reacting finely ground rock with concentrated phosphoric acid
- TSP contains 44-52% P_2O_5 . Common TSP in the market is 0-46-0
- Solubility: TSP is fully water-soluble
- TSP is most concentrated form of P-fertilizer and therefore, most cost-effective

Mode of application

- Apply TSP in bands at planting close to seed rows as for SSP
- Similar to SSP in other agronomic properties

8.2.3 Potassium fertilizers

1. Muriate of Potash (KCl) 0-0-60 (MOP)

- MOP is free flowing without handling and storage problems
- Anti-caking agents eliminate caking of MOP under high humidity
- Completely soluble in water
- On application to the soil, K^+ is adsorbed and retained by soil colloids. This minimizes possibility of leaching losses. From soil colloids it enters the soil solution for uptake by plant roots
- Entire requirement of MOP may be applied as basal dose. But split application is recommended in sandy soils, high rainfall areas and wetland rice fields
- MOP is neutral in reaction. This makes it suitable for most crops and soils. It is not suitable for chloride sensitive crops like tobacco and potatoes. Chloride lowers their quality

Mode of application

The K fertilizers are generally applied at sowing time with few exceptions where leaching losses can be expected. The adsorption of K^+ by soil colloid makes them remain available yet largely protected against leaching. The utilization rate of K is only 50 – 60% during the first year of application.

8.2.4 Complete and Incomplete Fertilizers

Complete and incomplete fertilizers are classified as ammonium phosphates, and nitro-phosphates and NPK fertilizers. They generally have high nutrient content.

1. Ammonium phosphates

The common ones are monoammonium phosphates (MAP) and diammonium phosphates (DAP). MAP contains 52-55% P_2O_5 that is almost entirely water soluble and 11-12% N. MAP currently available in the Kenyan market is of 11-52-0. It's non-hygroscopic and compatible with many other fertilizer materials. It is applied in planting holes or incorporated when applied to growing crops. DAP analysis is 18-46-0 and has similar properties to MAP.

Agronomic aspects of MAP and DAP

- Ammonium phosphates have desirable properties of NH_4^+ -N and water soluble P_2O_5
- DAP is suitable in areas and crops where P requirement is relatively higher than N
- DAP has an acidifying effect requiring lime application
- MAP has a wide N: P_2O_5 ratio and may require additional nitrogen

2. Nitro phosphates and NPK fertilizers

- Are formed when phosphate rock is acidulated with nitric acid or a mixture of nitric acid and phosphoric or sulphuric acids followed by ammoniation of the slurry. The slurry is then granulated or prilled with addition of N and K to give the required NPK grade
- Nitrophosphates are coated to minimize moisture being picked up
- Agronomic performance of nitrophosphates depends on solubility of the phosphate. Those that are at least 60% water-soluble are applicable to all crops and soils
- The less water soluble ones can be only for acidic soils and long duration crops e.g. tea and coffee
- Depending on production process, 30-50% N is in the NO_3^- and the rest in the NH_4^+ forms
- Examples are 23-23-0, 20-20-0, 20-10-10, 17-17-17, and 15-15-15

Mode of application

- Nitrophosphates are best applied as basal dressings

Complete Fertilizers

Complete fertilizers are those that contain all three primary nutrients: Nitrogen (N), Phosphorus (P), and Potassium (K) in varying proportions. Sometimes called NPK fertilizers. They are designed to supply multiple nutrients simultaneously for crops with balanced nutrient requirements.

Examples

17:17:17 → equal proportions of N, P, K

20:10:10 → more nitrogen, moderate phosphorus, lower potassium

12:12:17 + micronutrients → N, P, K with added secondary or micronutrients

Advantages

Supplies multiple nutrients in one application.

Simplifies fertilizer management.

Reduces labor and application costs.

Ensures balanced nutrition for crops, improving yield and quality.

Limitations

May not match specific soil nutrient needs exactly.

Risk of over-application of some nutrients.

Often more expensive than straight fertilizers.

8.3 Fertilizer Blending

Fertilizer blending refers to dry mixing of fertilizer granules without chemical reaction to produce a multi-nutrient fertilizer of a required nutrient content. Blended fertilizers are cheaper to produce than complex and compound fertilizers. They are made from mixing basic fertilizers such as ammonium nitrate, urea, calcium ammonium nitrate, di-ammonium phosphate, and natural materials such as rock phosphates.



Figure 8.1: Fertilizer blends sold for planting maize and compound 17.17.17 for coffee respectively.

Examples of blended fertilizers are those made by Athi River Mining Company in Kenya labelled Mavuno fertilizers. The company blends specific fertilizers for wheat, bananas, sugar cane, tobacco, coffee and tea. The fertilizer comes in two forms of planting and top dressing. Unlike most of the fertilizers currently in the Kenyan market, Mavuno fertilizer is blended to supply the primary nutrients to the soil in the required proportions. The fertilizer is composed of the following major nutrients, Nitrogen, phosphorus, Sulphur, Potassium Calcium and Magnesium plus micronutrients (minor) nutrients Zinc, Copper, Molybdenum, Boron and Manganese

Mavuno fertilizers are available in various packages i.e. 1kg, 10kg, 25kg and 50kg. Thus the farmers are able to buy what they can afford depending on the size of the farm.



Figure 8.2: Blended fertilizer for specific crops: green vegetables, fruits and vegetables and pulses and root vegetables

Advantages of fertilizer blending

1. Fertilizer blending provides a means for growers to receive the correct ratio of N P K and micronutrients. With increased use of soil analysis and scientific crop management, fertilizer blending is often the best way to give the grower exactly what his crops requires at specific time.
2. Blending is cheaper than fertilizers manufacture; hence countries without primary fertilizer industries are able to produce fertilizers that meet local farmers' needs.
3. Through blending, fertilizer suppliers can meet almost any analysis requirements quickly and efficiently.

Disadvantages of fertilizer blending

1. Some blended fertilizers may not be compatible with each other, producing undesirable effects such as chemical reaction, stickiness and/or caking (which may lead to handling difficulties).
2. Segregation or separation of individual constituents during handling, transportation or application leading to different parts of a field getting different rates of N P or K.
3. Once opened, blended fertilizes tend to cake during storage due to differences in hygroscopic nature (tendency to absorb moisture from the air) of some of the constituents.

Overcoming disadvantages of blended fertilizers

- Guidelines have been developed to assist fertilizer blenders, (for instance, Guidelines for the Compatibility of Fertilizer Blending Materials, developed by European Fertilizer Manufacturers' Association, based in Brussels, Belgium).
- State of the art computerized equipment have been developed for blending fertilizer (for instance, High Integrity Mixers for bulk blending with micronutrients, with capacity to add liquid coatings on granules)

- Farmers are advised to purchase required quantities and avoid storage once the bags have been opened.

How to choose what fertilizer to use and when

When choosing what fertilizer to use and how much, one needs to consider the following five factors:

- **Type of crop and its nutrient requirements:** some crops require a supply of certain nutrients more than others e.g. fruit trees will require a lot of K while leafy vegetables require more of N
- **Stage of growth of crop** - At planting plants require more of phosphorus for root development
- **Soil reaction** – some fertilizers like DAP or Urea should not be used in highly acidic soils as this would make some nutrients unavailable for uptake by plants
- **Soil analysis results** – This is the best guide on what type and amount of fertilizer to use
- **Weather condition** - when it is very dry and crops are water stressed, foliar feed is the best option. Top dressing is recommended only when the soil is moist

8.5 Environmental effects of fertilizer use

First accusations linking mineral fertilizers with the deterioration of the environment were made in early 1970s when leaching of nitrate was considered to lower farming profitability. Worldwide, the average share of nitrogen is therefore twice as high as necessary – with catastrophic effects for the environment, soil fertility and the climate (see Chapter 7). Given this oversupply, a significant portion is washed out in the form of nitrates, or escapes as nitrous oxide gas into the atmosphere.

There are several environmental effects of inorganic fertilizers on the environment which include:

a) Groundwater contamination

1. Nitrate leaching:

- All arable soils contain nitrogen in varying amounts, but on average 3,000-5,000 kg of N/ha is present in the plough layer. About 95% of this is bound in organic matter. Microbial activities in soil transform organic N into plant available form. This **mineralization process can free up to 500 kg/ha of N annually**; estimates place it at about 40-50 kg/ha on average. Nitrate leaching happens even when no N fertilizers are applied.
- Excess application of nitrogen fertilizer results in high leaching.
- High nitrate levels in drinking water is harmful to human health, especially infants and pregnant women. In babies it causes “Blue baby syndrome”; Nitrate

is non-toxic but it is reduced to nitrites by bacteria and when ingested, causes the blue baby syndrome in babies.

b) Greenhouse gaseous nitrogen emissions

- **Carbon dioxide emissions:** The manufacturing process of nitrogen fertilizers, such as ammonia production from natural gas or Haber-Bosch synthesis, is energy-intensive and emits significant amounts of CO₂. Similarly, the transportation of fertilizers to agricultural fields also contributes to CO₂ emissions.
- **Nitrous oxide (N₂O) emissions:** Nitrogen fertilizers, particularly those containing ammonium or urea, can undergo biochemical processes in the soil that result in the release of nitrous oxide (N₂O), a potent greenhouse gas. Today, global agriculture accounts for around 12 % of all greenhouse gas emissions. It thus ranks among the prime originators of climate change. Of this amount, about half (47 %) are caused by forest clearance and burning to clear land for farming. Another 17% (or one-third of the remaining agricultural emissions) are attributed to nitrous oxide released into the air as a result of nitrogen fertilization. Nitrogen brings about enormous short-term yield increases but is, at the same time, harmful to the soil and climate
- **Ammonia (NH₃) Volatilization:** Ammonium-based fertilizers, such as urea and ammonium nitrate, can undergo volatilization, releasing ammonia (NH₃) gas into the atmosphere. Ammonia emissions can contribute to air pollution, smog formation, and acid deposition, leading to environmental and human health concerns.

C) Eutrophication

Eutrophication is a process that occurs when water bodies become overly enriched with nutrients, particularly nitrogen (N) and phosphorus (P), leading to excessive growth of algae and aquatic plants. The excessive growth of alga and their decomposition disrupts aquatic ecosystems by depleting oxygen. Oxygen depletion causes harm to aquatic organisms such as fish and other aquatic animals. Eutrophication can be prevented by using slow-release fertilizers, soil conservation to reduce run off, and applying the right amount of fertilizer at the right time..

d) Cadmium

Cadmium is considered to be the most dangerous pollutant of agricultural soils. In most phosphate rock deposits, there are various levels of cadmium and other harmful elements that are tied with rock material and cannot be removed from ground rock mechanically. Various methods have been developed to remove cadmium but they all make use of phosphate uneconomical. Many industrialized countries have regulated or recommended maximum levels of cadmium in fertilizers.

e) Loss of Biodiversity: Intensive use of inorganic fertilizers can alter soil microbial communities and reduce soil biodiversity, which play crucial roles in nutrient cycling, soil health, and ecosystem functioning. Changes in soil microbial diversity and activity can have cascading effects on plant-microbe interactions, nutrient availability, and overall ecosystem stability

f) Soil acidification: By far the largest part of synthetic nitrogen fertilizers is based on ammonia. This comprises urea, ammonium nitrate, ammonium sulfate, and ammonium phosphate. All of them accelerate soil acidification. Although nitrogen (N) fertilizers themselves are not inherently acidic, their application to soil can lead to acidification over time. This phenomenon is primarily due to the nitrification process, whereby soil microbes convert ammonium nitrogen (NH_4^+) from fertilizers into nitrate nitrogen (NO_3^-). During this process, hydrogen ions (H^+) are released, contributing to soil acidification.

Additionally, excessive application of nitrogen fertilizers can disrupt the balance of soil nutrients, leading to increased leaching of cations such as calcium (Ca^{2+}), magnesium (Mg^{2+}), and potassium (K^+), which are replaced by hydrogen ions, further exacerbating soil acidity.

Over the long term, continuous use of nitrogen fertilizers without proper soil management practices can result in soil acidification, negatively impacting soil health, nutrient availability, and overall crop productivity. Therefore, it is essential to monitor soil pH levels and implement appropriate liming or soil amendment strategies to mitigate the adverse effects of acidification caused by nitrogen fertilizer applications.

Mitigation Measures

To minimize environmental harm, farmers should adopt 4R Nutrient Stewardship:

1. **Right source** – choose the proper type of fertilizer for the soil and crop.
2. **Right rate** – apply only the required amount.
3. **Right time** – apply when the crop can use it most efficiently.
4. **Right place** – position fertilizer close to the root zone to reduce losses.

Additional measures include:

- Using **split applications** of N fertilizers.
- Use of slow release fertilizers
- Incorporating fertilizers into soil instead of surface broadcasting.
- Planting **cover crops** to reduce leaching.
- Employing **buffer strips** near water bodies.
- Promoting **integrated nutrient management (INM)** by combining mineral fertilizers with organic inputs.

CHAPTER 9: ORGANIC FERTILIZERS

Learning outcomes

By the end of this lesson, you will be able to:

1. Define organic fertilizers and explain their importance in crop production.
2. Identify and classify different types of organic fertilizers (e.g., farmyard manure, compost, green manure, crop residues, and animal wastes).
3. Describe the nutrient content and properties of various organic fertilizers.
4. Explain how organic fertilizers improve soil fertility, structure, and microbial activity.
5. Discuss the advantages and limitations of using organic fertilizers compared to mineral fertilizers.

Overview

In Chapter 8, you learned about **inorganic/mineral fertilizers**. Chapter 9 is about the other form of fertilizers ie **organic fertilizers**. Organic fertilizers are natural materials of either plant or animal origin, including livestock manure, green manures, crop residues, household waste, compost, and woodland litter. They are important in soil fertility management for supplying nutrients but also for improving soil health through contributing to soil organic matter.

Organic fertilizers are natural materials of either plant or animal origin, including livestock manure, green manures, crop residues, household waste, compost, and woodland litter.

9.1 Role of organic fertilizers

Organic fertilizers or inputs can influence nutrient availability by:

1. Increasing the total nutrients added (ADD NUTRIENTS)
2. Controlling the net mineralization-immobilization patterns (SYNERGY)
3. Acting as a source of carbon and energy to drive microbial activities (ENERGY SOURCE)
4. Contributing to formation of organic matter which have positive effects on crop growth such as increasing porosity, infiltration, resistance to erosion and ease of root penetration. (SOM FORMATION)
5. Improve soil structure and general soil health
6. Through interactions with the mineral therefore reducing toxic cations and the P sorption capacity of the soil: INCREASE P AVAILABILITY

In addition to these direct effects on nutrient availability, organic materials can affect, root growth, pests and soil physical properties.

9.2 Farmer-available organic materials in Africa

1. Animal/Farmyard Manure

Livestock manure: from Cattle, shoats, poultry, pigs etc. This is available on most farms where there are livestock but the quality is often low due to poor methods of preparation and utilization. Good manure management is necessary to obtain high quality manure that will improve crop productivity

For good quality boma manure, the following factors are important:

- Livestock boma should be roofed to avoid leaching of valuable nutrients into the ground due to rains
- Sufficient bedding to absorb the urine should be provided to ensure that the boma is not wet. This will avoid losses of nitrogen to the air. Loss of nitrogen to the air is indicated by a strong smell of ammonia around the cattle boma
- After removing the manure from the boma, it should be kept covered with grass so that it can mature. This reduces losses of N through volatilisation
- When it is taken to the field, it should be placed in furrows or planting holes and covered with soil to minimize nutrient losses

2. Crop residues

Crop residues consists of the non harvested portions of crop plants and are readily available to small scale farmers as an organic resource. They are usually low in nutrients and high in lignin. This limits their use as an organic resource for soil fertility improvement. They need to be applied appropriately because inappropriate application may result in poor crop performance eg when fresh residues of maize stover is incorporated at the start of the season. Crop residues also have competing uses eg livestock feed and fuel

3. Compost manure

- Compost making is the process of turning organic material into humus.
- Naturally humus improves soil water holding capacity, soil structure, and is a source of plant nutrients.
- Compost can be made easily at the farm using crop residues, garden weeds, kitchen and household wastes, hedge cuttings and any other vegetative material. Incorporating plants/weeds rich in certain nutrients enriches the compost with the specific nutrients, e.g., tithonia, calliandra, leucaena enriches with N.
- Compost should be made at a location close to the place where it will be used.

4. Agroforestry options

a. Biomass transfer using Agroforestry Trees

- Biomass transfer or cut and carry systems involve growing of trees and/or shrubs along contours on farms or on roadsides and then consequently spreading their leaves on crop fields at planting time or later in the season (as green manure)
- *Tithonia diversifolia* (tithonia), *Calliandra calothyrsus* (calliandra) and *Leucaena trichandra* (leucaena) are some of the trees used in biomass transfer systems
- These shrubs also withstand frequent cutting/coppicing producing new re-growth within a short time of 2 weeks especially under warm and wet conditions
- The practice of using *Tithonia diversifolia* (tithonia), *Calliandra calothyrsus* (calliandra) and *Leucaena trichandra* (leucaena) to improve soil fertility is recent following research

findings that have shown that application of these organic materials into the soil increases soil productivity.

- These agroforestry trees are also important for enhancing nutrient cycling in farming systems.
- Biomass or prunings harvested is chopped and incorporated in to the soil about one week before planting the crop

Tithonia diversifolia

Kiruru (Kikuyu); Kirurite (Kiembu); Ciamuganani (Gichuka)

- Tithonia or Mexican sunflower is a succulent shrub that originated from Mexico.
- It is a common roadside weed where it grows to up to a height of 1 to 3 meters.
- One major advantage of tithonia is that it produces large quantities of biomass and tolerates regular pruning, which is a favourable characteristic for its use for biomass transfer. Compared to many other plants, the biomass contains high levels of nutrients
- When tithonia is applied into the soil it decomposes rapidly releasing these nutrients that are readily available for plant uptake.
- Farmers commonly plant tithonia as hedges especially along the homesteads. This is evident in central Kenya districts such as Embu, Meru, and western Kenya, Vihiga, Maseno, Malawi
- These hedges are maintained through cutting and in most cases the biomass is either left on the ground to rot or thrown into the cow-sheds.
- Tithonia growing along the roadsides when it overgrows is cut by the road maintenance personnel and in most cases left at the site. All this material is lost yet it can be utilized for improving soil fertility.

Calliandra calothyrsus

Common name (“kariandra”)

- Calliandra is a small leguminous tree belonging to the family Mimosoideae that originated from Central America.
- Calliandra grows well in the tropics at altitudes ranging from sea level to 1900 m above sea level (asl) and where the average rainfall is above 1000 mm.
- It is among the best nitrogen fixing trees, and nodulates very well with nitrogen fixing bacteria (rhizobia).
- Prunings harvested from calliandra contain substantial amounts of nutrients in the leaves; averaging 3.5% nitrogen, 1.6% potassium, 0.21% phosphorus, 0.6% calcium, and 0.24% magnesium.
- It is good fodder and can either be incorporated directly into the soil or fed to livestock and then recycle manure

Leucaena trichandra

Common name (“Rukina”)

- *Leucaena trichandra* is generally a new species in Kenya that was recently identified to be resistant to leucaena psyllid insect pest (*Heterophylla cubana*)

- *L. trichandra* is a leguminous shrub belonging to the family mimosoidea. It has a strong, deep root system that allows it to tolerate drought once established.
- It grows well in lowland dry to humid tropics (below 500 m asl, 600 -1700 mm annual rainfall).
- It is deep rooted, and nodulates well with rhizobia fixing large quantities of N in their biomass. It tolerates regular cutting and coppices well.
- The leaves are rich in nutrients and contain on average 3.7% nitrogen, 3.3 % potassium, 0.26% phosphorus, 1.2% calcium and 0.3% magnesium. The leaves decompose faster than those of *Calliandra* due to the lower tannin content in *Leucaena*.

b. Improved fallows

Improved fallows is a rotational system that uses preferred fallow tree species as opposed to colonization by natural vegetation like in shifting cultivation tree species restore soil fertility quicker & shorten fallow period.

- *(The fallow system is an old practice where cropped fields are left to lie fallow to allow the soil to rest, auto-regenerate and restore soil fertility. Traditionally, these fallows were used for grazing or were left to natural vegetation but when improved fallow agroforestry system is applied the trees replace crops on the fallow fields in sequence over time).*
- ❖ However, due to increased population pressure, the fallow periods have reduced and are no longer effective in maintaining soil fertility. This has led to development of improved tree fallows that have shown great potential as an agro-forestry technique especially in the densely populated areas of western Kenya where there is bimodal rainfall, allowing farmers to fallow for one season per year.
- ❖ *Crotalaria grahamiana* and *Tephrosia vogelii* are some of the most popular species with farmers in western Kenya and seeds are broadcast or planted directly. The trees are planted at the time of second weeding of maize field and then left to occupy the field alone in the following rainy season, which is usually the short season

Short-term fallows of leguminous trees and herbaceous cover crops can provide a practical means of nutrient cycling when grown in rotation with cereal crops. Two-year tree fallows of *Sesbania* [*Sesbania sesban*] (L) Merr.] or *Tephrosia* (*Tephrosia vogelii* Hook. F.) have replenished soil N levels enough to grow three subsequent high-yielding maize crops in N-depleted

5. Herbaceous legumes

- Legumes enrich the soil by biological atmospheric nitrogen fixation and can replace nitrogen fertilizers, wholly or in part.
- Until the advent mineral fertilizers, legumes and animal manures were also used widely through North America and Europe.
- While mineral fertilizers have displaced legume pasture legumes and green manuring in industrialized countries, legumes still contribute significantly to soil productivity in developing countries especially in the tropical and subtropical regions of the world where the nitrogen fixation capacity of legumes can be exploited more fully throughout the year.

- Legumes directly contribute to soil nitrogen supply and thereby have a sparing effect on the amount of additional fertilizers required for high cereal yields.
- An example is *Mucuna pruriens* which derives an average of 70% of its N from atmospheric nitrogen.
- Other are dolichos (*Lablab purpureus*), and sun hemp (*Crotalaria ochroleuca*), nodulation and nitrogen fixation, ground cover, pests and disease resistance
- They can either be intercropped or grown in rotation

Disadvantages of organic fertilizers

1. Low nutrient concentration compared to mineral fertilizers, requiring large quantities.
2. Slow nutrient release, which may not meet immediate crop demands.
3. Bulky and difficult to transport and store.
4. Variable nutrient content, depending on source and decomposition stage.
5. May contain weed seeds or pathogens if not properly composted.
6. Odor problems, especially with fresh manures.
7. Nutrient losses can occur during storage, composting, or application.
8. Labour-intensive to apply compared to concentrated mineral fertilizers.
9. May require large land areas to produce sufficient quantities for high-yielding crops.

CHAPTER 10: INTEGRATED SOIL FERTILITY MANAGEMENT

By the end of this lesson, students will be able to:

1. Define Integrated Soil Fertility Management (ISFM) and explain its significance in sustainable agriculture.
2. Identify and describe the main components of ISFM, including organic inputs, mineral fertilizers, and improved crop varieties.
3. Discuss the role of soil management practices (e.g., crop rotation, cover crops, and conservation tillage) in ISFM.
4. Evaluate the challenges of implementing ISFM in smallholder and commercial farming systems.
5. Apply ISFM principles to design a soil fertility improvement plan for a specific crop or farm

One of the biggest challenges in the tropics is developing soil fertility management strategies that address farmers' resource constraints, increase food production, reduce risk, and enhance soil fertility.

Research has developed recommendations on soil fertility management, but these recommendations have not benefited the farmers due to low adoption. The reasons for low adoption are many and complex.

The situation of soil fertility problem in SSA is complex because of three main reasons;

1. **African soils are amongst the poorest of the world.** Compared to other continents, a large proportion of soils in SSA have low soil fertility and exhibit a variety of constraints, including low nutrient availability, low OM, moisture stress, and high erodibility. The most dominant constraint is the poor organic matter (OM) status (quality and quantity) and the inherent low nitrogen (N) availability, closely followed by low phosphorus (P) availability. In extended regions, the situation is still worsened by high soil acidity.
2. **Small holder farmers have complex farming systems**, that are heterogeneous and difficult to develop recommendations that work for all
3. **Poverty levels limit application of mineral fertilizers:** Recommendations on rates of fertilizer use have been developed but farmers are not able to afford adequate amounts

An integrated approach is needed to deal with this complexity. Integrated Soil Fertility Management (ISFM) was coined that involve a holistic approach

Integrated Soil Fertility Management (ISFM) is a holistic approach that combines **organic and inorganic nutrient sources** with **improved crop and soil management practices** to maximize crop productivity sustainably.

ISFM combines a mix of organic and inorganic materials, used with close attention to timing and placing of the inputs to maximise nutrient use efficiency.

What is ISFM?

ISFM is defined as *the use of mineral fertilizer, organic inputs, and improved germplasm combined with the knowledge on how to adapt these practices to local conditions*. The objective is to optimize agronomic use efficiency of the applied nutrients and improving crop productivity. All inputs need to be managed following sound agronomic and economic principles. The interventions under ISFM are aimed at improving soil health conditions and optimal crop response to nutrients in a sustainable and economic way. use efficiency and crop productivity. These practices necessarily include appropriate fertilizer and organic input management in combination with utilization of improved germplasm

9.3 Paradigm shifts in soil fertility management

- ISFM is a paradigm shift in the wake of numerous technological advances, which have not, in spite of concerted efforts, been widely used by smallholder farms due to the **complexity of farming systems**.
- Paradigm shifts in soil fertility research (Table 1) in an effort to improve adoption of developed innovations

Table 1: Paradigm shifts to address the problem of soil fertility

Period	Problem	Paradigm
1950's -1960's	Farmers are ignorant of technical solutions	Top down approaches • Linear approach: Research → Extension → Farmer • Each seen as a discrete function feeding onto the other • Farmers' role: Learn, Adopt, Conform • Farmers seen as Early adopters; slow/progressive adopters; or Laggards
1970 -mid 80's	Farmers have constraints and can contribute to solutions	Scientists trying to understand farmers; Farmer participation is important and surveys were conducted to generate information Books written: Farmer-back-to-Farmer (Rhoades and Booth 1982), Farmer First & last (Chambers and Ghildyal, 1985), Farmer first (Chambers, 1989)
1990's	Improve information exchange	Farmer participatory research and development (learning, dissemination adoption); Farmer involvement in problem diagnosis, experimentation, testing; Integrated soil fertility management
From 2000	Change institutions and Policy	Emphasis is on multi-stakeholder linkages Different versions but quite similar conceptually & objectively – Integrated soil fertility management – IAR4D (Integrated Agricultural Research 4 development (Co-generation of knowledge, processes, innovation) – Innovation Systems approach (Facilitators create innovation platforms – Value chain analysis (VCA)

ISFM is an approach (a way of doing things) that incorporates an integration of strategies/options for managing soil fertility for sustainable agricultural production and food security

8.4 Approaches and Practices of ISFM

8.4.1 What are we integrating?

1. Integrated use of mineral fertilizer and organic materials. This is integrated nutrient management, which is the technical backbone of ISFM. It involves judicious manipulation of nutrient input, outputs and internal flows to achieve productive and sustainable agricultural systems
2. Integrating the beneficial and deleterious effects of abiotic factors, such as water harvesting and soil conservation methods, to control soil erosion and improve water capture and use efficiency.
3. Integration of legumes with multi –purpose roles in agroecosystems
4. Crop-livestock integration to improve the sustainability of nutrient cycles
5. Integration and greater use of local and indigenous knowledge, innovations, practices and resources, and science knowledge–based–management systems. Integration of farmers' and scientists' knowledge
6. Integration of policy and institutional framework, as well as on-site and off-site effects
7. Integration and judicious management of interactions between soil fertility and biotic stresses such as crop diseases, pests, and weeds

8.4.2 Integrated soil management principles

In integrated soil management, practices that increase nutrient recycling, use of organic sources of plant nutrients and improve fertilizer use efficiency should receive greater attention. Examples of these are;

1. **Improved crop rotations:** legumes grown as a food crop or live mulch (cover crop) can be successfully rotated with a crop that produces high biomass or intercropped with tree species (e.g. alley cropping) in order to provide N, enhance organic matter content and prevent soil erosion;
2. **Maximising organic matter production:** through green manure, cover crops and agroforestry;
3. **Enhancing nutrient cycling :** (soil micro-organisms, plant roots, predator insects) through managing plant-soil-pest-predator interactions; soil cover (mulch, cover crops) to supply nutrients, reduce weeds and labour, reduce heat and nutrient losses and enhance functions of soil biota and plant roots;
4. **No or reduced tillage:**, Conservation agricultural practices that in addition to the benefits of improving soil health and enhanced productivity. Also reduces labour and energy costs while enhancing water quality, soil conservation and carbon sequestration;
5. **Maximizing biological Nitrogen Fixation:** Practices for this include selection and breeding of crop species with higher nitrogen use efficiency, resilience to deficiencies of P ,and nitrogen-fixing capacity; and inoculation
6. **Maximising crop, soil and animal biodiversity:** to reduce disease and pest outbreaks and maintain a balanced nutrient supply, provide resilience and multiple benefits (through OM decomposition, nutrient cycling and enhanced soil structure).

9.4.3 ISFM Technologies

National and international Agricultural organizations increasingly use ISFM approaches; Documented ISFM technologies include;

1. **Integrated nutrient management systems:**
2. **Intercropping systems:** cereal legume intercropping systems eg maize and common beans
3. **Integration of multipurpose legumes** (green manure, cover crops and herbaceous, tree, forage or fodder and grain or food legumes)
4. **Improved manure management:**
5. **Improved fallows:**
6. **Microdose use of fertilizers**
7. **Biomass transfer** of high-quality organic inputs

Use of regionally and locally available inorganic sources of plant nutrients and amendments (rock phosphate, limestone).